

A Proof of the Exact Convergence Rate of Gradient Descent*

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Abstract

We prove the exact worst-case convergence rate of gradient descent for smooth strongly convex optimization on $\mathbb{R}^d$. Concretely, assuming that the objective function f is μ -strongly convex and L -smooth, we identify the smallest possible value of τ for which the inequality $f(x_N) - f_* \leq \tau \|x_0 - x_*\|^2$ always holds. The result was previously conjectured by Drori and Teboulle for the case $\mu = 0$, and by Taylor, Hendrickx, and Glineur for the case $\mu > 0$.

1 Introduction

We study the problem of minimizing a differentiable function f on $\mathbb{R}^d$, that is,

$$\min_{x \in \mathbb{R}^d} f(x). \quad (1.1)$$

The algorithm we consider is gradient descent (GD) with a fixed stepsize $\gamma > 0$. Given an initial point $x_0 \in \mathbb{R}^d$, the algorithm updates the iterates as

$$x_{k+1} = x_k - \gamma \nabla f(x_k) \quad \text{for } k = 0, 1, \dots \quad (\text{GD})$$

In this paper, we answer the following open question in convex optimization: “How quickly does gradient descent converge to the minimum?” This informal question will be mathematically formulated in the next two subsections.

1.1 Assumptions

In this subsection, we summarize the standard assumptions which we will follow throughout the paper. We assume that f has a minimizer x_* , which may not be unique. When dealing with quantities depending on x_* , we consider x_* to be chosen arbitrarily among them. We denote $f_* = f(x_*)$.

It is common to assume that f is convex in order to make the problem (1.1) tractable. To ensure that GD with a predetermined fixed stepsize always converges, it is usually assumed that the gradient of f is Lipschitz continuous with a known Lipschitz constant. For our purposes, the following working definitions suffice to cover these two assumptions: Given parameters $-\infty < \mu < L < \infty$, we assume that

- f is μ -strongly convex, that is, for all $p, q \in \mathbb{R}^d$, we have

$$0 \geq f(p) - f(q) + \langle \nabla f(p), q - p \rangle + \frac{\mu}{2} \|p - q\|^2; \quad (1.2)$$

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- f is L -smooth, that is, for all $p, q \in \mathbb{R}^d$, we have

$$0 \leq f(p) - f(q) + \langle \nabla f(p), q - p \rangle + \frac{L}{2} \|p - q\|^2. \quad (1.3)$$

Throughout the paper, we always assume $L > 0$. Depending on the context, we sometimes require $\mu \geq 0$. Keep in mind that we allow $\mu < 0$ unless specified otherwise. When $\mu = 0$, the notion of μ -strong convexity is equivalent to the standard notion of convexity. When f is convex, it is known that the L -smoothness of f is equivalent to the L -Lipschitz continuity of ∇f (see, for example, [12, Thm. 2.1.5]). We denote by $\mathcal{F}_{\mu,L}$ the space of μ -strongly convex and L -smooth functions on $\mathbb{R}^d$ which have a minimizer. The ratio μ/L is denoted by κ . The following property is well-known (see, for example, [12, Thm. 2.1.5], [5, Thm. A.3], and [14, Thm. 3.2]).

Proposition 1.1. *When $f \in \mathcal{F}_{\mu,L}$, the interpolation inequality*

$$\begin{aligned} 0 \geq & f(p) - f(q) + \langle \nabla f(p), q - p \rangle \\ & + \frac{1}{2L} \|\nabla f(p) - \nabla f(q)\|^2 + \frac{\mu L}{2(L - \mu)} \left\| p - q - \frac{1}{L} (\nabla f(p) - \nabla f(q)) \right\|^2 \end{aligned} \quad (1.4)$$

holds for all $p, q \in \mathbb{R}^d$.

1.2 History of the problem

The study of the convergence properties of gradient descent has a long history. The method of gradient descent can be traced back to [2, 8]. An early work on the convergence of gradient descent is [3], where neither the convexity of f nor the Lipschitz continuity of ∇f is assumed. As a result, line search (i.e., finding an appropriate stepsize $\gamma > 0$ at each iteration) was necessary, and the rate of convergence was not shown. In the following decades, the convergence properties of GD became well-studied as the standard assumptions in Section 1.1 were adopted. In particular, it was shown in [13, Thm. 6] that when $f \in \mathcal{F}_{\mu,L}$ with $\mu \geq 0$, the iterates of GD with $\gamma \in (0, 2/L)$ satisfy the inequality

$$\|x_N - x_*\| \leq \max\{|1 - \gamma\mu|, |1 - \gamma L|\}^N \|x_0 - x_*\|. \quad (1.5)$$

When $\mu > 0$, this inequality not only shows that $x_N \rightarrow x_*$ as $N \rightarrow \infty$, but also provides an upper bound on the rate at which the distance $\|x_N - x_*\|$ decreases. We refer to the ratio which we aim to bound above ($\|x_N - x_*\|/\|x_0 - x_*\|$ in this case) as the *performance criterion*.

From a practical perspective, it is more reasonable to measure the quality of x_N as an approximate solution for the optimization problem (1.1) using $f(x_N) - f_*$ rather than $\|x_N - x_*\|$.¹ For example, the following bounds can be found in [12, §2.1.5] (see also [1, §3] and [5, §4]).

Proposition 1.2. *When $f \in \mathcal{F}_{0,L}$, the iterates of GD with $\gamma = 1/L$ satisfy*

$$f(x_N) - f_* \leq \frac{2L}{N + 4} \|x_0 - x_*\|^2. \quad (1.6)$$

¹Another reason to consider the performance criterion $(f(x_N) - f_*)/\|x_0 - x_*\|^2$ is that when $\mu = 0$, one cannot prove meaningful bounds for the performance criteria $\|x_N - x_*\|/\|x_0 - x_*\|$ and $(f(x_N) - f_*)/(f(x_0) - f_*)$. In those cases, $\|x_N - x_*\| \leq \|x_0 - x_*\|$ and $f(x_N) - f_* \leq f(x_0) - f_*$ are the best possible bounds (see [16]), indicating that GD yields no improvement in these respects.

When $f \in \mathcal{F}_{\mu,L}$ with $\mu \geq 0$, the iterates of GD with $\gamma = 2/(\mu + L)$ satisfy

$$f(x_N) - f_* \leq \frac{L}{2} \left(\frac{L - \mu}{L + \mu} \right)^{2N} \|x_0 - x_*\|^2. \quad (1.7)$$

Here, the performance criterion is set as $(f(x_N) - f_*)/\|x_0 - x_*\|^2$, which is perhaps the most widely studied performance criterion in the convex optimization literature. A natural question is whether the inequalities (1.6) and (1.7) can be tightened. This leads to the problem of identifying the value

$$\sup \left\{ \frac{f(x_N) - f_*}{\|x_0 - x_*\|^2} : f \in \mathcal{F}_{\mu,L}, x_0 \in \mathbb{R}^d, x_k \text{ generated by GD} \right\} \quad (1.8)$$

for given γ, N, μ , and L . Since $f(x_k)$ may not converge when $\gamma \notin (0, 2/L)$,² we assume $\gamma \in (0, 2/L)$. Based on numerical evidence, a closed-form expression of the value (1.8) was conjectured in [4] for the case $\mu = 0$. Later, this conjecture was generalized in [15] to encompass the case $\mu > 0$. The conjectured result is shown in the following theorem, which we will prove.

Theorem 1.3 ([4, Conj. 1], [15, Conj. 2]). *When $f \in \mathcal{F}_{\mu,L}$ with $\mu \geq 0$, the iterates of GD with $\gamma \in (0, 2/L)$ satisfy*

$$f(x_N) - f_* \leq \max \left\{ \frac{1}{1 + \gamma L \sum_{j=1}^{2N} (1 - \gamma\mu)^{-j}}, (1 - \gamma L)^{2N} \right\} \frac{L \|x_0 - x_*\|^2}{2}. \quad (1.9)$$

Note that Theorem 1.3 only provides an upper bound of (1.8). To identify the value (1.8), we also need the matching lower bound. This is done by the following proposition.

Proposition 1.4 ([4], [15, §4.1.2]). *There is a pair $(f, x_0) \in \mathcal{F}_{\mu,L} \times \mathbb{R}^d$ for which (1.9) holds with equality.*

To the best of our knowledge, the upper bound (Theorem 1.3) has only been partially proved for the case where $\mu = 0$ and $\gamma \in (0, 1/L)$ in [4, Thm. 2] and [17, Thm. 2], while the lower bound (Proposition 1.4) was completely proved in [4] for the case $\mu = 0$,³ and in [15] for the case $\mu > 0$.

Although the connection to Theorem 1.3 was not clear, a recent work [14] made notable progress by proving the exact convergence rate of GD for another performance criterion $\|\nabla f(x_N)\|^2/(f(x_0) - f_*)$. As in the previous case, both the upper and lower bounds are needed. Their results are shown below.

Theorem 1.5 ([14, Thm. 2.2]). *When $f \in \mathcal{F}_{\mu,L}$ with $\mu \geq 0$, the iterates of GD with $\gamma \in (0, 2/L)$ satisfy*

$$\frac{\|\nabla f(x_N)\|^2}{2L} \leq \max \left\{ \frac{1}{1 + \gamma L \sum_{j=1}^{2N} (1 - \gamma\mu)^{-j}}, (1 - \gamma L)^{2N} \right\} (f(x_0) - f_*). \quad (1.10)$$

Proposition 1.6 ([14, Prop. 5.2]). *There is a pair $(f, x_0) \in \mathcal{F}_{\mu,L} \times \mathbb{R}^d$ for which (1.10) holds with equality.*

²Consider the objective function $f(x) = (L/2)\|x\|^2$, which is in $\mathcal{F}_{\mu,L}$.

³See Theorem 3.2 of the preprint version ([arXiv:1206.3209v1](https://arxiv.org/abs/1206.3209v1)) of their paper.

In another recent work [7], it was conjectured that Theorem 1.3 for the optimal stepsize $\gamma = \gamma^*(N, \mu, L)$ (see Section 4.2) can be proved using the PEP methodology (see Section 1.3) with a structured set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$. Although we prove Theorem 3.1 in a different way from the conjectured one, our proof and their conjectured proof share similarities. See Remark 5.2 for details.

1.3 Previously known partial proof

In this subsection, we review the partial proof of Theorem 1.3 presented in [4], which is based on the methodology of *performance estimation problem* (PEP).⁴ It is assumed that $\mu = 0$.

Introduce the index set $I_N^* := \{0, \dots, N, *\}$. Suppose that there is a set of nonnegative *multipliers* $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ such that the inequality

$$\begin{aligned} f_N - f_* - \max \left\{ \frac{1}{1 + 2N\gamma L}, (1 - \gamma L)^{2N} \right\} \frac{L \|x_0 - x_*\|^2}{2} \\ \leq \sum_{i,j \in I_N^*} \lambda_{i,j} \left(f_j - f_i + \langle g_j, x_i - x_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2 \right) \end{aligned} \quad (1.11)$$

holds for all [families $\{x_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and $x_{k+1} = x_k - \gamma g_k$ for $k = 0, \dots, N-1$]. Then, we can put $x_i \leftarrow x_i$,⁵ $g_i \leftarrow \nabla f(x_i)$, and $f_i \leftarrow f(x_i)$ for all $i \in I_N^*$, and then the right-hand side of (1.11) is nonpositive by Proposition 1.1. Thus, the inequality (1.11) implies the desired inequality (1.9).

Therefore, to prove Theorem 1.3, it suffices to set appropriate multipliers $\lambda_{i,j}$ and then verify that the inequality (1.11) holds under the stated conditions. An advantage of this approach is that the task of finding working multipliers $\lambda_{i,j}$ can be formulated as a semidefinite program, and thus one can use numerical solvers to confirm their existence and even infer their closed-form expressions. A difficulty that arises here is that feasible solutions $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ to this semidefinite program are not unique. In other words, there is some freedom in choosing $\{\lambda_{i,j}\}_{i,j \in I_N^*}$. This causes numerical solvers to yield inconsistent solutions. To resolve this issue (although they did not explicitly mention the reason), the authors of [4] imposed an additional constraint

$$\lambda_{i,j} = 0 \quad \text{for all } (i,j) \notin \{(k, k+1), (*, k)\}_{0 \leq k \leq N-1} \cup \{(*, N)\}. \quad (1.12)$$

This constraint removes the freedom in selecting $\{\lambda_{i,j}\}_{i,j \in I_N^*}$, enabling them to infer a closed-form expression of the solution $\{\lambda_{i,j}\}_{i,j \in I_N^*}$. This strategy was successful for the case $\gamma \in (0, 1/L]$, leading to a partial proof of Theorem 1.3. However, they were not able to extend their proof to the case $\gamma \in (1/L, 2/L)$, since feasible solutions $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfying (1.12) do not always exist in this case (this is suggested by the numerical evidence in [15, Table 1]).

⁴By definition, a performance estimation problem (PEP) is an optimization problem of finding the value (1.8) (or its variants). The analysis in this subsection was originally derived in [4] by taking the Lagrangian dual of a certain relaxation of PEP. We will refer to any convergence analysis involving the verification of inequalities like (1.11) as a PEP analysis.

⁵We abused notation here. The left-hand x_k refer to arbitrary points in the previous sentence, while the right-hand x_k denote the iterates of GD and the given minimizer x_* .

1.4 Our strategy

In this paper, we prove Theorem 1.3 for any strong convexity parameter $\mu \in [0, L)$ and any stepsize $\gamma \in (0, 2/L)$. We use the PEP methodology in the previous subsection, but even for the case where $\mu = 0$ and $\gamma \in (0, 1/L]$, we choose a different set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ from the one used in [4].

A crucial ingredient of our proof is observing the phenomenon that some convergence analyses for a first-order optimization algorithm $\mathbf{H}$ with respect to the performance criterion $(f(x_N) - f_*)/\|x_0 - x_*\|^2$ have *dual* convergence analyses for the *anti-transposed* algorithm $\mathbf{H}^A$ with respect to the performance criterion $\|\nabla f(x_N)\|^2/(f(x_0) - f_*)$. See [11, Thm. 3] and [10, Thms. 1,2] for a recent theory on this phenomenon. To compare our approach for obtaining a proof of Theorem 1.3 with the one in [4], we handle the nonuniqueness of feasible solutions $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ by removing the freedom in their choice within the dual analysis.⁶

The structure of our paper is outlined as follows. In Section 3, we develop a theory for the aforementioned phenomenon. Our theory encompasses the existing theory and is supported by a new proof. In this correspondence, proving Theorem 1.5 will mirror the task of proving Theorem 1.3. In Section 4, we provide a proof of Theorem 1.5, which differs from the previous one presented in [14]. In Section 5, we combine the two results to complete the proof of Theorem 1.3.

Our proof involves tedious symbolic computations. Rather than presenting written calculations, we provide MATLAB codes for verifying the computations. The codes are available at the following GitHub repository:

<https://github.com/jungbinkim1/GD-Exact-Rate>.

2 Preliminaries and notation

We mostly follow the notations in [14]. We define $\rho = 1 - \gamma L$ and $\eta = 1 - \gamma \mu$. For $k = 1, 2, \dots$, we define functions $E_k(\cdot)$, $F_k(\cdot)$, and $T_k(\cdot, \cdot)$ as

$$\begin{aligned} E_k(x) &= \begin{cases} \sum_{j=1}^{2k} x^{-j} & \text{if } x \neq 0, \\ \infty & \text{if } x = 0, \end{cases} \\ F_k(x) &= \sum_{j=1}^k x^j, \\ T_k(\rho, \eta) &= E_k(\eta) - E_k(\rho). \end{aligned}$$

See Table 1 for a rough description of the function E_k . We introduce the index sets $I_N = \{0, \dots, N\}$ and $I_N^* = I_N \cup \{*\}$. For convenience, we set all multipliers $\lambda_{i,j}$ (or $\nu_{i,j}$) with $i = j$ to zero throughout the paper.

We denote the space of N -tuples of points in $\mathbb{R}^d$ by $(\mathbb{R}^d)^N$. We represent elements of this space as column vectors, that is, $[p_1, \dots, p_N]^T$ with $p_k \in \mathbb{R}^d$ for $k = 1, \dots, N$. To point out trivial facts, we note that the space $(\mathbb{R}^d)^N$ is equipped with the inner product $\langle [p_1, \dots, p_N]^T, [q_1, \dots, q_N]^T \rangle = \sum_{k=1}^N \langle p_k, q_k \rangle$, and that $N \times N$ matrices can be viewed as operators on $(\mathbb{R}^d)^N$.

⁶We empirically observed that our proof strategy fully removes the freedom in choosing the multipliers. We will not prove this because it is not essential to our main objective.

Table 1: The behavior of the the function E_k for any fixed k .

x	-1	$\cdots$	0	$\cdots$	∞
$E_k(x)$	0	$\nearrow$	∞	$\searrow$	0

2.1 Properties of interpolation inequality

The following proposition shows that the interpolation inequality (1.4) tightens as μ increases or L decreases.

Proposition 2.1. *Fix $p, q, \nabla f(p), \nabla f(q), f(p)$, and $f(q)$. Denote the right-hand side of (1.4) as $Q(\mu, L)$. Then, we have*

- (i) $Q(\mu, L) \geq Q(\mu', L)$ whenever $\mu > \mu'$;
- (ii) $Q(\mu, L) \geq Q(\mu, L')$ whenever $L < L'$.

Proof. These follow from

$$\begin{aligned}\frac{\partial Q}{\partial \mu} &= \frac{L^2}{2(L-\mu)^2} \left\| p - q - \frac{1}{L} (\nabla f(p) - \nabla f(q)) \right\|^2 \geq 0, \\ \frac{\partial Q}{\partial L} &= -\frac{\mu^2}{2(L-\mu)^2} \left\| (p - q) - \frac{1}{\mu} (\nabla f(p) - \nabla f(q)) \right\|^2 \leq 0.\end{aligned}$$

□

Since $\nabla f(x_*) = 0$, the inequality (1.4) with $p = x_*$ can be written as

$$0 \geq f_* - f(q) + \frac{1}{2L} \|\nabla f(q)\|^2 + \frac{\mu L}{2(L-\mu)} \left\| x_* - q + \frac{1}{L} \nabla f(q) \right\|^2.$$

Thus, when $\mu \geq 0$, we have $f(q) - \frac{1}{2L} \|\nabla f(q)\|^2 \geq f_*$. The following proposition shows that this inequality remains valid even when $\mu < 0$.

Proposition 2.2. *If f is L -smooth, then the inequality $f(x) - \frac{1}{2L} \|\nabla f(x)\|^2 \geq f_*$ holds for all $x \in \mathbb{R}^d$.*

Proof. By putting $p = x$ and $q = x - \frac{1}{L} \nabla f(x)$ in the inequality (1.3), we obtain $f(x - \frac{1}{L} \nabla f(x)) \leq f(x) - \frac{1}{2L} \|\nabla f(x)\|^2$. Combine this with $f_* \leq f(x - \frac{1}{L} \nabla f(x))$. □

2.2 Optimiation algorithms as lower-triangular matrices

Consider first-order methods where each iterate x_k is in $x_0 + \text{span}\{\nabla f(x_0), \dots, \nabla f(x_{k-1})\}$. For such a method, there is an $N \times N$ lower-triangular matrix $\mathbf{H}$ such that

$$\begin{bmatrix} x_1 - x_0 \\ \vdots \\ x_N - x_{N-1} \end{bmatrix} = -\frac{1}{L} \begin{bmatrix} \mathbf{H}_{1,1} & & \\ \vdots & \ddots & \\ \mathbf{H}_{N,1} & \cdots & \mathbf{H}_{N,N} \end{bmatrix} \begin{bmatrix} g_0 \\ \vdots \\ g_{N-1} \end{bmatrix}, \quad (2.1)$$

where $g_k = \nabla f(x_k)$ for $k = 0, \dots, N-1$. In particular, GD can be expressed as (2.1) with $\mathbf{H} = (\gamma L)\mathbf{I}_N$, where $\mathbf{I}_N$ is the $N \times N$ identity matrix. Conversely, any lower-triangular matrix $\mathbf{H}$ gives an implementable algorithm (2.1). Therefore, we can identify the algorithm (2.1) with the lower-triangular matrix $\mathbf{H}$.

We can reparametrize (2.1) so that the iterates on the left-hand side become $x_k^+ = x_k - \frac{1}{L}g_k$ instead of x_k . Define an $(N+1) \times (N+1)$ matrix $\tilde{\mathbf{H}}$ as

$$\tilde{\mathbf{H}} = \begin{bmatrix} 1 & & & & \\ \mathbf{H}_{1,1} - 1 & 1 & & & \\ \vdots & \vdots & \ddots & & \\ \mathbf{H}_{N-1,1} & \mathbf{H}_{N-1,2} & \cdots & 1 & \\ \mathbf{H}_{N,1} & \mathbf{H}_{N,2} & \cdots & \mathbf{H}_{N,N} - 1 & 1 \end{bmatrix}. \quad (2.2)$$

Then, (2.1) can be equivalently written as

$$\mathbf{x} = -\frac{1}{L}\tilde{\mathbf{H}}\mathbf{g}, \quad (2.3)$$

where $\mathbf{x} = [x_0^+ - x_0, x_1^+ - x_0^+, \dots, x_N^+ - x_{N-1}^+]^T$ and $\mathbf{g} = [g_0, \dots, g_N]^T$.

2.3 Anti-transpose of optimization algorithm

For an $N \times N$ matrix $\mathbf{H}$, its *anti-diagonal transpose* (or shortly *anti-transpose*) $\mathbf{H}^A$ is obtained by reflecting $\mathbf{H}$ over its anti-diagonal, that is, $\mathbf{H}_{i,j}^A = \mathbf{H}_{N+1-j, N+1-i}$ for $i, j = 1, \dots, N$. When $\mathbf{H}$ is lower-triangular, the first-order algorithm associated with $\mathbf{H}^A$ can be written as

$$\begin{bmatrix} y_1 - y_0 \\ \vdots \\ y_N - y_{N-1} \end{bmatrix} = -\frac{1}{L} \begin{bmatrix} \mathbf{H}_{N,N} & & \\ \vdots & \ddots & \\ \mathbf{H}_{N,1} & \cdots & \mathbf{H}_{1,1} \end{bmatrix} \begin{bmatrix} g_0 \\ \vdots \\ g_{N-1} \end{bmatrix}, \quad (2.4)$$

where $g_k = \nabla f(y_k)$ for $k = 0, \dots, N-1$. We denoted the iterates by y_k instead of x_k to prevent future confusion. In particular, the anti-transpose of the matrix $(\gamma L)\mathbf{I}_N$ is itself. Perhaps, the observation of the anti-transpose relationship between first-order optimization algorithms dates back to [9, §6.6]. Note that $\widetilde{\mathbf{H}^A} = \tilde{\mathbf{H}}^A$ holds, where the tilde in the notation is defined as in (2.2). Thus, with $y_k^+ = y_k - \frac{1}{L}g_k$, (2.4) can be equivalently written as

$$\mathbf{y} = -\frac{1}{L}\tilde{\mathbf{H}}^A\mathbf{g}, \quad (2.5)$$

where $\mathbf{y} = [y_0^+ - y_0, y_1^+ - y_0^+, \dots, y_N^+ - y_{N-1}^+]^T$ and $\mathbf{g} = [g_0, \dots, g_N]^T$.

The anti-transpose can be defined using the language of operators as follows.

Definition 2.3. Let $\mathbf{H}$ be an operator on $(\mathbb{R}^d)^N$. Then, its anti-transpose is the operator $\mathbf{H}^A$ on $(\mathbb{R}^d)^N$ that satisfies

$$\langle \mathbf{H}x, \mathbf{T}y \rangle = \langle \mathbf{T}x, \mathbf{H}^A y \rangle$$

for all $x, y \in (\mathbb{R}^d)^N$, where $\mathbf{T}$ is the time reversal operator on $(\mathbb{R}^d)^N$ defined as

$$\mathbf{T}([x_1, \dots, x_N]^T) = [x_N, \dots, x_1]^T.$$

Then, by definition, we have $H^A = TH^T T$ (this identity can be found in [6, §1]). When H is invertible, from the identity $(H^T)^{-1} = (H^{-1})^T$, it follows that

$$(H^A)^{-1} = (TH^T T)^{-1} = (T)^{-1} (H^T)^{-1} (T)^{-1} = T (H^{-1})^T T = (H^{-1})^A.$$

This justifies denoting this operator as H^{-A} .

3 Equivalence between convergence analyses for minimizing $(f(x_N) - f_*)/\|x_0 - x_*\|^2$ and minimizing $\|\nabla f(x_N)\|^2/(f(x_0) - f_*)$

In this section, we show that there is an equivalence between some PEP analyses for the algorithm $\mathbf{H}$ with the performance criterion $(f(x_N) - f_*)/\|x_0 - x_*\|^2$ and some PEP analyses for the algorithm $\mathbf{H}^A$ with the performance criterion $\|\nabla f(x_N)\|^2/(f(x_0) - f_*)$.

3.1 PEP analysis for performance criterion $(f(x_N) - f_*)/\|x_0 - x_*\|^2$

In this subsection, we present a PEP analysis aimed at proving convergence guarantees in the form $f(x_N) - f_* \leq \frac{\tau}{2}\|x_0 - x_*\|^2$. Our analysis differs from the one in [4] by using $\{x_{k+1}^+ - x_k^+\}$ instead of $\{g_k\}$ as the basis for the quadratic form,⁷ although both analyses aim at the same performance criterion.

Theorem 3.1. *Given a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$, the inequality*

$$\begin{aligned} f_N - f_* - \frac{\tau}{2} \|x_0 - x_*\|^2 \\ \leq \sum_{i,j \in I_N^*} \lambda_{i,j} \left[f_j - f_i + \langle g_j, x_i - x_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2 \right. \\ \left. + \frac{\mu L}{2(L - \mu)} \left\| x_i - x_j - \frac{1}{L} (g_i - g_j) \right\|^2 \right] \end{aligned} \quad (3.1)$$

holds for all [families $\{x_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.1)] if and only if the following two conditions are satisfied:

(F1)

$$\sum_{i \in I_N^*} \lambda_{i,k} - \sum_{j \in I_N^*} \lambda_{k,j} = \begin{cases} 1 & \text{if } k = N, \\ -1 & \text{if } k = *, \\ 0 & \text{otherwise.} \end{cases}$$

(F2) For any points $x_0, x_0^+, \dots, x_N^+, x_*^+, y_0, y_0^+, \dots, y_N^+, y_*^+ \in \mathbb{R}^d$ satisfying

$$\begin{aligned} y_k^+ - y_{k-1}^+ &= \sum_{i \in I_N^*} \lambda_{i,N-k} (x_i^+ - x_{N-k}^+) \quad \text{for } k = 1, \dots, N, \\ \frac{y_0^+ + y_*^+}{2} - y_N^+ &= \frac{x_N^+ - x_*^+}{2} + \sum_{i \in I_N^*} \lambda_{i,*} (x_i^+ - x_*^+), \end{aligned} \quad (3.2)$$

⁷This will become clearer in Section 4.1, where we present a PEP analysis in terms of the positive semidefiniteness of a certain matrix.

the quantity

$$\begin{aligned}
S = & -L \left\langle \tilde{\mathbf{H}}^{-1} \mathbf{x}, \mathbf{T} \mathbf{y} \right\rangle + \frac{L}{2} \left\| y_0 - \frac{y_0^+ + y_*^+}{2} - \frac{x_N^+ - x_*^+}{2} \right\|^2 \\
& - \frac{\mu L}{L - \mu} \left[\sum_{j=1}^N \left\langle y_j^+ - y_{j-1}^+, x_{N-j}^+ \right\rangle \right. \\
& \quad \left. + \left\langle \frac{y_0^+ + y_*^+}{2} - y_N^+, x_*^+ \right\rangle + \left\langle \frac{y_0^+ - y_*^+}{2}, x_N^+ \right\rangle \right],
\end{aligned} \tag{3.3}$$

where

$$\begin{aligned}
\mathbf{x} &= [x_0^+ - x_0, x_1^+ - x_0^+, \dots, x_N^+ - x_{N-1}^+]^T, \\
\mathbf{y} &= [y_0^+ - y_0, y_1^+ - y_0^+, \dots, y_N^+ - y_{N-1}^+]^T,
\end{aligned}$$

is nonnegative.

Proof. Note that the inequality (3.1) holds for all [families $\{x_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.1)] if and only if the inequality

$$\begin{aligned}
& f_N^+ - f_*^+ + \frac{1}{2L} \|g_N\|^2 - \frac{\tau}{2} \|x_0 - x_*^+\|^2 \\
& \leq \sum_{i,j \in I_N^*} \lambda_{ij} \left(f_j^+ - f_i^+ + \langle g_j, x_i^+ - x_j^+ \rangle + \frac{\mu L}{2(L - \mu)} \|x_i^+ - x_j^+\|^2 \right).
\end{aligned} \tag{3.4}$$

holds for all [families $\{x_0\} \cup \{x_i^+\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i^+\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.3)].⁸

For (3.4) to hold for any $\{f_k^+\}_{k \in I_N^*}$, the terms f_k^+ in (3.4) should vanish, which is (F1). When (F1) holds, we can rewrite (3.4) using $\mathbf{g} = -L\tilde{\mathbf{H}}^{-1}\mathbf{x}$ as

$$\begin{aligned}
0 & \leq S_1(x_0, x_0^+, \dots, x_N^+, x_*^+) \\
& := \frac{\tau}{2} \|x_0 - x_*^+\|^2 - \frac{L}{2} \|(\tilde{\mathbf{H}}^{-1}\mathbf{x})_{N+1}\|^2 \\
& \quad - L \left\langle \tilde{\mathbf{H}}^{-1}\mathbf{x}, \begin{bmatrix} \sum_{i \in I_N^*} \lambda_{i,0} (x_i^+ - x_0^+) \\ \sum_{i \in I_N^*} \lambda_{i,1} (x_i^+ - x_1^+) \\ \vdots \\ \sum_{i \in I_N^*} \lambda_{i,N} (x_i^+ - x_N^+) \end{bmatrix} \right\rangle + \frac{\mu L}{2(L - \mu)} \sum_{i,j \in I_N^*} \lambda_{ij} \|x_i^+ - x_j^+\|^2,
\end{aligned}$$

where $(\tilde{\mathbf{H}}^{-1}\mathbf{x})_{N+1}$ denotes the $(N+1)$ -th entry of $\tilde{\mathbf{H}}^{-1}\mathbf{x}$. Thus, showing the inequality (3.4) is equivalent to showing $\inf S_1 \geq 0$, where the infimum is taken over all points $x_0, x_0^+, \dots, x_N^+, x_*^+$ in

⁸To show $(\Rightarrow)$, put $x_i = x_i^+ + \frac{1}{L}g_i$ and $f_i = f_i^+ + \frac{1}{2L}\|g_i\|^2$ for all $i \in I_N^*$. To show $(\Leftarrow)$, put $x_i^+ = x_i - \frac{1}{L}g_i$ and $f_i^+ = f_i - \frac{1}{2L}\|g_i\|^2$ for all $i \in I_N^*$.

$\mathbb{R}^d$. We introduce *dual* iterates $y_0^+, \dots, y_N^+, y_*^+$ by the rule (3.2). Using (F1) and (3.2), we have

$$\begin{aligned}
& \sum_{i \in I_N^*} \lambda_{i,N} (x_i^+ - x_N^+) \\
&= \sum_{i,j \in I_N^*} \lambda_{i,j} (x_i^+ - x_j^+) - \sum_{j \in I_N^* \setminus \{N\}} \sum_{i \in I_N^*} \lambda_{i,j} (x_i^+ - x_j^+) \\
&= (x_*^+ - x_N^+) - \sum_{k=1}^N (y_k^+ - y_{k-1}^+) - \left(\frac{y_0^+ + y_*^+}{2} - y_N^+ - \frac{x_N^+ - x_*^+}{2} \right) \\
&= y_0^+ - v,
\end{aligned} \tag{3.5}$$

where $v = \frac{y_0^+ + y_*^+}{2} + \frac{x_N^+ - x_*^+}{2}$. We can now rewrite (3.4) as

$$\begin{aligned}
0 &\leq S_2(x_0, x_0^+, \dots, x_N^+, x_*^+, y_0^+, \dots, y_N^+, y_*^+) \\
&:= \frac{\tau}{2} \|x_0 - x_*^+\|^2 - \frac{L}{2} \|(\tilde{\mathbf{H}}^{-1} \mathbf{x})_{N+1}\|^2 \\
&\quad - L \left\langle \tilde{\mathbf{H}}^{-1} \mathbf{x}, \begin{bmatrix} y_N^+ - y_{N-1}^+ \\ \vdots \\ y_1^+ - y_0^+ \\ y_0^+ - v \end{bmatrix} \right\rangle + \frac{\mu L}{2(L-\mu)} \sum_{i,j \in I_N^*} \lambda_{ij} \|x_i^+ - x_j^+\|^2.
\end{aligned}$$

Thus, showing (3.4) is equivalent to showing $\inf S_2 \geq 0$, where the infimum is taken over all points $x_0, x_0^+, \dots, x_N^+, x_*^+, y_0^+, \dots, y_N^+, y_*^+$ in $\mathbb{R}^d$ satisfying (3.2). Introduce an auxiliary point $y_0 \in \mathbb{R}^d$ and consider the quantity

$$\begin{aligned}
& S_3(x_0, x_0^+, \dots, x_N^+, x_*^+, y_0, y_0^+, \dots, y_N^+, y_*^+) \\
&:= \frac{L}{2} \|y_0 - v\|^2 + \frac{\tau}{2} \|x_0 - x_*^+\|^2 \\
&\quad - L \left\langle \tilde{\mathbf{H}}^{-1} \mathbf{x}, \mathbf{T} \mathbf{y} \right\rangle + \frac{\mu L}{2(L-\mu)} \sum_{i,j \in I_N^*} \lambda_{ij} \|x_i^+ - x_j^+\|^2.
\end{aligned}$$

When the other points are fixed, one can verify that $S_2 = \inf_{y_0} S_3$. Thus, showing (3.4) is equivalent to showing $\inf S_3 \geq 0$, where the infimum is taken over all points $x_0, x_0^+, \dots, x_N^+, x_*^+, y_0, y_0^+, \dots, y_N^+, y_*^+$ in $\mathbb{R}^d$ satisfying (3.2). We proceed by rewriting the last term of S_3 as follows:

$$\begin{aligned}
& \sum_{i,j \in I_N^*} \lambda_{i,j} \|x_i^+ - x_j^+\|^2 \\
&= \sum_{k \in I_N^*} \left(\sum_{j \in I_N^*} \lambda_{k,j} + \sum_{i \in I_N^*} \lambda_{i,k} \right) \|x_k^+\|^2 - 2 \sum_{j \in I_N^*} \sum_{i \in I_N^*} \lambda_{i,j} \langle x_i^+, x_j^+ \rangle \\
&= \sum_{k \in I_N^*} \left(\sum_{j \in I_N^*} \lambda_{k,j} - \sum_{i \in I_N^*} \lambda_{i,k} \right) \|x_k^+\|^2 - 2 \sum_{j \in I_N^*} \left\langle \sum_{i \in I_N^*} \lambda_{i,j} (x_i^+ - x_j^+), x_j^+ \right\rangle.
\end{aligned}$$

We simplify the first term using (F1), and the second term using (3.2) and (3.5).

$$\begin{aligned}
& \sum_{i,j \in I_N^*} \lambda_{i,j} \|x_i^+ - x_j^+\|^2 \\
&= -\|x_N^+\|^2 + \|x_*^+\|^2 - 2 \sum_{k=1}^N \langle y_k^+ - y_{k-1}^+, x_{N-k}^+ \rangle \\
&\quad - 2 \left\langle \frac{y_0^+ + y_*^+}{2} - y_N^+ - \frac{x_N^+ - x_*^+}{2}, x_*^+ \right\rangle - 2 \left\langle \frac{y_0^+ - y_*^+}{2} - \frac{x_N^+ - x_*^+}{2}, x_N^+ \right\rangle \\
&= -2 \left(\sum_{k=1}^N \langle y_k^+ - y_{k-1}^+, x_{N-k}^+ \rangle + \left\langle \frac{y_0^+ + y_*^+}{2} - y_N^+, x_*^+ \right\rangle + \left\langle \frac{y_0^+ - y_*^+}{2}, x_N^+ \right\rangle \right).
\end{aligned}$$

This completes the proof. $\square$

3.2 PEP analysis for performance criterion $\|\nabla f(x_N)\|^2/(f(x_0) - f_*)$

In this subsection, we present a PEP analysis aimed at proving convergence guarantees in the form $\frac{1}{2\tau} \|\nabla f(y_N)\|^2 \leq f(y_0) - f_*$. Our analysis differs from the one in [9] by using $\{y_{k+1}^+ - y_k^+\}$ instead of $\{g_k\}$ as the basis for the quadratic form (see Footnote 7), although both analyses aim at the same performance criterion.

Theorem 3.2. *Given a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$, the inequality*

$$\begin{aligned}
& \frac{1}{2\tau} \|g_N\|^2 - f_0 + f_* - \frac{L}{2} \left\| \sum_{i \in I_N^*} \nu_{i,*} \left(y_i - \frac{1}{L} g_i - y_* \right) \right\|^2 \\
& \leq \sum_{i,j \in I_N^*} \nu_{i,j} \left[f_j - f_i + \langle g_j, y_i - y_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2 \right. \\
& \quad \left. + \frac{\mu L}{2(L-\mu)} \left\| y_i - y_j - \frac{1}{L} (g_i - g_j) \right\|^2 \right]
\end{aligned} \tag{3.6}$$

holds for all [families $\{y_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.4)] if and only if the following two conditions are satisfied:

(G1)

$$\sum_{i \in I_N^*} \nu_{i,k} - \sum_{j \in I_N^*} \nu_{k,j} = \begin{cases} 1 & \text{if } k = *, \\ -1 & \text{if } k = 0, \\ 0 & \text{otherwise.} \end{cases}$$

(G2) For any points $x_0, x_0^+, \dots, x_N^+, x_*^+, y_0, y_0^+, \dots, y_N^+, y_*^+ \in \mathbb{R}^d$ satisfying

$$\begin{aligned} x_k^+ - x_{k-1}^+ &= \sum_{i \in I_N^*} \nu_{i, N-k} (y_i^+ - y_{N-k}^+) \quad \text{for } k = 1, \dots, N-1, \\ x_0^+ - x_*^+ &= \sum_{i \in I_N^*} \nu_{i, N} (y_i^+ - y_N^+), \\ \frac{x_*^+ - x_N^+}{2} &= \frac{y_*^+ - y_0^+}{2} + \sum_{i \in I_N^*} \nu_{i, *} (y_i^+ - y_*^+), \end{aligned} \tag{3.7}$$

the quantity (3.3) is nonnegative.

Proof. Note that the inequality (3.6) holds for all [families $\{y_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.4)] if and only if the inequality

$$\begin{aligned} &\frac{1}{2\tau} \|g_N\|^2 + f_*^+ - f_0^+ - \frac{1}{2L} \|g_0\|^2 - \frac{L}{2} \left\| \sum_{i \in I_N^*} \nu_{i, *} (y_i^+ - y_*^+) \right\|^2 \\ &\leq \sum_{i, j \in I_N^*} \nu_{ij} \left(f_j^+ - f_i^+ + \langle g_j, y_i^+ - y_j^+ \rangle + \frac{\mu L}{2(L - \mu)} \|y_i^+ - y_j^+\|^2 \right) \end{aligned} \tag{3.8}$$

holds for all [families $\{y_0\} \cup \{y_i^+\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i^+\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.5)] (see Footnote 8).

For (3.8) to hold for any $\{f_k^+\}_{k \in I_N^*}$, the terms f_k^+ in (3.8) should vanish, which is (G1). When (G1) holds, we can rewrite (3.8) using $\mathbf{g} = -L\tilde{\mathbf{H}}^{-A}\mathbf{y}$ (in particular, we have $g_0 = -L(y_0^+ - y_0)$ since $\tilde{\mathbf{H}}_{1,1}^{-A} = 1$) as

$$\begin{aligned} 0 &\leq S_1(y_0, y_0^+, \dots, y_N^+, y_*^+) \\ &:= \frac{L}{2} \|y_0^+ - y_0\|^2 - \frac{L^2}{2\tau} \left\| (\tilde{\mathbf{H}}^{-A}\mathbf{y})_{N+1} \right\|^2 + \frac{L}{2} \left\| \sum_{i \in I_N^*} \nu_{i, *} (y_i^+ - y_*^+) \right\|^2 \\ &\quad - L \left\langle \tilde{\mathbf{H}}^{-A}\mathbf{y}, \begin{bmatrix} \sum_{i \in I_N^*} \nu_{i, 0} (y_i^+ - y_0^+) \\ \sum_{i \in I_N^*} \nu_{i, 1} (y_i^+ - y_1^+) \\ \vdots \\ \sum_{i \in I_N^*} \nu_{i, N} (y_i^+ - y_N^+) \end{bmatrix} \right\rangle + \frac{\mu L}{2(L - \mu)} \sum_{i, j \in I_N^*} \nu_{ij} \|y_i^+ - y_j^+\|^2, \end{aligned}$$

where $(\tilde{\mathbf{H}}^{-A}\mathbf{y})_{N+1}$ denotes the $(N+1)$ -th entry of $\tilde{\mathbf{H}}^{-A}\mathbf{y}$. Thus, showing the inequality (3.8) is equivalent to showing $\inf S_1 \geq 0$, where the infimum is taken over all points $y_0, y_0^+, \dots, y_N^+, y_*^+$ in

$\mathbb{R}^d$. We introduce dual iterates $x_0^+, \dots, x_N^+, x_*^+$ by the rule (3.7). Using (G1) and (3.7), we have

$$\begin{aligned}
\sum_{i \in I_N^*} \nu_{i,0} (y_i^+ - y_0^+) &= \sum_{i,j \in I_N^*} \nu_{i,j} (y_i^+ - y_j^+) - \sum_{j \in I_N^* \setminus \{0\}} \sum_{i \in I_N^*} \nu_{i,j} (y_i^+ - y_j^+) \\
&= (y_0^+ - y_*^+) - \sum_{k=1}^{N-1} (x_k^+ - x_{k-1}^+) \\
&\quad - (x_0^+ - x_*^+) - \left(\frac{x_*^+ - x_N^+}{2} - \frac{y_*^+ - y_0^+}{2} \right) \\
&= x_N^+ - x_{N-1}^+ + y_0^+ - v,
\end{aligned} \tag{3.9}$$

where $v = \frac{y_0^+ + y_*^+}{2} + \frac{x_N^+ - x_*^+}{2}$. Note that the last equality in (3.7) can be written as $\sum_{i \in I_N^*} \nu_{i,*} (y_i^+ - y_*^+) = y_0^+ - v$. We can now rewrite (3.8) as

$$\begin{aligned}
0 &\leq S_2(x_0^+, \dots, x_N^+, x_*, y_0, y_0^+, \dots, y_N^+, y_*) \\
&:= \frac{L}{2} \|y_0^+ - y_0\|^2 - \frac{L^2}{2\tau} \left\| (\tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y})_{N+1} \right\|^2 + \frac{L}{2} \|y_0^+ - v\|^2 \\
&\quad - L \left\langle \tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y}, \begin{bmatrix} x_N^+ - x_{N-1}^+ + y_0^+ - v \\ \vdots \\ x_1^+ - x_0^+ \\ x_0^+ - x_*^+ \end{bmatrix} \right\rangle + \frac{\mu L}{2(L-\mu)} \sum_{i,j \in I_N^*} \nu_{ij} \|y_i^+ - y_j^+\|^2 \\
&= \frac{L}{2} \|y_0 - v\|^2 - \frac{L^2}{2\tau} \left\| (\tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y})_{N+1} \right\|^2 \\
&\quad - L \left\langle \tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y}, \begin{bmatrix} x_N^+ - x_{N-1}^+ \\ \vdots \\ x_1^+ - x_0^+ \\ x_0^+ - x_*^+ \end{bmatrix} \right\rangle + \frac{\mu L}{2(L-\mu)} \sum_{i,j \in I_N^*} \nu_{ij} \|y_i^+ - y_j^+\|^2.
\end{aligned}$$

Thus, showing (3.8) is equivalent to showing $\inf S_2 \geq 0$, where the infimum is taken over all points $x_0^+, \dots, x_N^+, x_*^+, y_0, y_0^+, \dots, y_N^+, y_*^+$ satisfying (3.7). Introduce an auxiliary point $x_0 \in \mathbb{R}^d$ and consider the quantity

$$\begin{aligned}
S_3(x_0, x_0^+, \dots, x_N^+, x_*^+, y_0, y_0^+, \dots, y_N^+, y_*^+) \\
&:= \frac{L}{2} \|y_0 - v\|^2 + \frac{\tau}{2} \|x_0 - x_*^+\|^2 \\
&\quad - L \left\langle \tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y}, \mathbf{T} \mathbf{x} \right\rangle + \frac{\mu L}{2(L-\mu)} \sum_{i,j \in I_N^*} \nu_{ij} \|y_i^+ - y_j^+\|^2.
\end{aligned}$$

When the other points are fixed, one can verify that $S_2 = \inf_{x_0} S_3$. Thus, showing (3.8) is equivalent to showing $S_3 \geq 0$, where the infimum is taken over all points $x_0, x_0^+, \dots, x_N^+, x_*, y_0, y_0^+, \dots, y_N^+, y_*$ satisfying (3.7). By Definition 2.3, we have $\langle \tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y}, \mathbf{T} \mathbf{x} \rangle = \langle \mathbf{T} \mathbf{y}, \tilde{\mathbf{H}}^{-1} \mathbf{x} \rangle$. We proceed by rewriting

the last term of S_3 as follows:

$$\begin{aligned}
& \sum_{i,j \in I_N^*} \nu_{i,j} \|y_i^+ - y_j^+\|^2 \\
&= \sum_{k \in I_N^*} \left(\sum_{j \in I_N^*} \nu_{k,j} + \sum_{i \in I_N^*} \nu_{i,k} \right) \|y_k^+\|^2 - 2 \sum_{j \in I_N^*} \sum_{i \in I_N^*} \nu_{i,j} \langle y_i^+, y_j^+ \rangle \\
&= \sum_{k \in I_N^*} \left(\sum_{j \in I_N^*} \nu_{k,j} - \sum_{i \in I_N^*} \nu_{i,k} \right) \|y_k^+\|^2 - 2 \sum_{j \in I_N^*} \left\langle \sum_{i \in I_N^*} \nu_{i,j} (y_i^+ - y_j^+), y_j^+ \right\rangle.
\end{aligned}$$

We simplify the first term using (G1), and the second term using (3.7) and (3.9).

$$\begin{aligned}
& \sum_{i,j \in I_N^*} \nu_{i,j} \|y_i^+ - y_j^+\|^2 \\
&= -\|y_*^+\|^2 + \|y_0^+\|^2 - 2 \sum_{k=1}^{N-1} \langle x_k^+ - x_{k-1}^+, y_{N-k}^+ \rangle - 2 \langle x_0^+ - x_*^+, y_N^+ \rangle \\
&\quad - 2 \left\langle \frac{x_*^+ - x_N^+}{2} - \frac{y_*^+ - y_0^+}{2}, y_*^+ \right\rangle - 2 \left\langle \frac{x_N^+ + x_*^+}{2} - x_{N-1}^+ + \frac{y_0^+ - y_*^+}{2}, y_0^+ \right\rangle \\
&= -2 \left(\sum_{k=1}^N \langle y_k^+ - y_{k-1}^+, x_{N-k}^+ \rangle + \left\langle \frac{y_0^+ + y_*^+}{2} - y_N^+, x_*^+ \right\rangle + \left\langle \frac{y_0^+ - y_*^+}{2}, x_N^+ \right\rangle \right).
\end{aligned}$$

This completes the proof. $\square$

The presence of the last term on the left-hand side of (3.6) is undesirable. The following theorem is a variant of Theorem 3.2, where this term is removed under an additional assumption that the point x_* is involved in the convergence analysis only through the interpolation inequality (1.4) with $p = x_*$ and $q = x_N$.

Theorem 3.3. *Given a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfying*

$$\nu_{i,*} = \begin{cases} 1 & \text{if } i = N, \\ 0 & \text{otherwise,} \end{cases} \quad \nu_{*,j} = 0 \quad \text{for } j = 0, \dots, N, \quad (3.10)$$

the inequality

$$\begin{aligned}
\frac{1}{2\tau} \|g_N\|^2 - f_0 + f_* &\leq \sum_{i,j \in I_N^*} \nu_{i,j} \left[f_j - f_i + \langle g_j, y_i - y_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2 \right. \\
&\quad \left. + \frac{\mu L}{2(L-\mu)} \left\| y_i - y_j - \frac{1}{L} (g_i - g_j) \right\|^2 \right] \quad (3.11)
\end{aligned}$$

holds for all families $\{y_i\}_{i \in I_N^}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.4) if and only if the two conditions (G1) and (G2) are satisfied.*

Proof. Assume that (3.10) holds. Then, the quantity S_1 in the original proof of Theorem 3.2 is the sum of $(\frac{L}{2} + \frac{\mu L}{2(L-\mu)})\|y_*^+ - y_N^+\|^2$ and other terms independent of y_*^+ , and is thus minimized at $y_*^+ = y_N^+$ when the other points $y_0, y_0^+, \dots, y_N^+$ are fixed.

If the inequality (3.11) is used instead of (3.6), mimicking the proof of Theorem 3.2 yields the following quantity instead of S_1 :

$$\begin{aligned} S'_1(y_0, y_0^+, \dots, y_N^+, y_*^+) &:= \frac{L}{2} \|y_0^+ - y_0\|^2 - \frac{L^2}{2\tau} \left\| (\tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y})_{N+1} \right\|^2 \\ &\quad - L \left\langle \tilde{\mathbf{H}}^{-\mathbf{A}} \mathbf{y}, \begin{bmatrix} \sum_{i \in I_N^*} \nu_{i,0} (y_i^+ - y_0^+) \\ \sum_{i \in I_N^*} \nu_{i,1} (y_i^+ - y_1^+) \\ \vdots \\ \sum_{i \in I_N^*} \nu_{i,N} (y_i^+ - y_N^+) \end{bmatrix} \right\rangle \\ &\quad + \frac{\mu L}{2(L-\mu)} \sum_{i,j \in I_N^*} \nu_{ij} \|y_i^+ - y_j^+\|^2. \end{aligned}$$

Since S'_1 is the sum of $\frac{\mu L}{2(L-\mu)}\|y_*^+ - y_N^+\|^2$ and other terms independent of y_*^+ , it is minimized at $y_*^+ = y_N^+$ when the other points $y_0, y_0^+, \dots, y_N^+$ are fixed.

Thus, we have

$$\begin{aligned} \inf_{y_0, y_0^+, \dots, y_N^+, y_*^+ \in \mathbb{R}^d} S_1(y_0, y_0^+, \dots, y_N^+, y_*^+) &= \inf_{y_0, y_0^+, \dots, y_N^+ \in \mathbb{R}^d} S_1(y_0, y_0^+, \dots, y_N^+, y_N^+) \\ &= \inf_{y_0, y_0^+, \dots, y_N^+ \in \mathbb{R}^d} S'_1(y_0, y_0^+, \dots, y_N^+, y_N^+) \\ &= \inf_{y_0, y_0^+, \dots, y_N^+, y_*^+ \in \mathbb{R}^d} S'_1(y_0, y_0^+, \dots, y_N^+, y_*^+). \end{aligned}$$

Therefore, showing $\inf S'_1 \geq 0$ is equivalent to showing $\inf S_1 \geq 0$. This means that Theorem 3.2 remains valid after replacing the inequality (3.6) by (3.11). $\square$

3.3 General correspondence between two PEP analyses

In this subsection, we present a technique to convert a PEP analysis using Theorem 3.2 (or Theorem 3.3) into a PEP analysis using Theorem 3.1, and vice versa. Suppose that we have a set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfying the two conditions (G1) and (G2), and that we want to find a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfying the two conditions (F1) and (F2). Then, we take the following two steps:

Step 1. For $j \neq N$, set the multipliers $\lambda_{i,j}$ so that (3.2) and (3.7) are equivalent.

Step 2. For $j = N$, set the multipliers $\lambda_{i,j}$ so that (F1) holds.

If these two steps are completed successfully, we can conclude that the two conditions (F1) and (F2) are satisfied for the resulting set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$.

For the opposite direction, we take the following two steps:

Step 1. For $j \neq 0$, set the multipliers $\nu_{i,j}$ so that (3.2) and (3.7) are equivalent.

Step 2. For $j = 0$, set the multipliers $\nu_{i,j}$ so that (G1) holds.

Remark 3.4. For a given set of non-negative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$, there might not exist a set of non-negative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ for which (3.2) and (3.7) are equivalent, and vice versa. Thus,

not all PEP analyses using Theorem 3.2 (or Theorem 3.3) can be translated to PEP analyses using Theorem 3.1, and vice versa.

3.4 Special correspondence between two relaxed PEP analyses

The results in this subsection will not be directly used in the rest of the paper, thus the reader can skip it. In this subsection, we show a full correspondence between every valid PEP analysis using Theorem 3.1 and every valid PEP analysis using Theorem 3.3, if it is further assumed that only a specific subset of interpolation inequalities can be used. Specifically, we *relax* the analysis in Section 3.1 by assuming (1.12), and the analysis in Section 3.2 by assuming

$$\nu_{i,j} = 0 \quad \text{for all } (i,j) \notin \{(k,k+1), (N,k)\}_{0 \leq k \leq N-1} \cup \{(N,*)\}. \quad (3.12)$$

By applying the technique in the previous subsection, we obtain the following lemma.

Lemma 3.5.

- (i) *If a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfies (G1) and (3.12), then there is a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfying (F1) and (1.12), such that (3.2) and (3.7) are equivalent.*
- (ii) *If a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfies (F1) and (1.12), then there is a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfying (G1) and (3.12), such that (3.2) and (3.7) are equivalent.*

Proof. (i) Let $A_k = 1 + \sum_{j=0}^{k-1} \nu_{N,j}$ for $k = 1, \dots, N$. Then, $\{A_k\}_{k=1}^N$ is a nondecreasing sequence in $[1, \infty)$. Using (3.12) and (G1), we can express all the multipliers $\nu_{i,j}$ in terms of A_k as

$$\nu_{i,j} = \begin{cases} A_j & \text{if } i = j-1, 1 \leq j \leq N, \\ A_1 - 1 & \text{if } i = N, j = 0, \\ A_{j+1} - A_j & \text{if } i = N, 1 \leq j \leq N-1, \\ 1 & \text{if } i = N, j = *, \\ 0 & \text{otherwise.} \end{cases} \quad (3.13)$$

Thus, (3.7) without the last equality can be written as

$$\begin{bmatrix} x_{N-1}^+ - x_{N-2}^+ \\ \vdots \\ x_1^+ - x_0^+ \\ x_0^+ - x_*^+ \end{bmatrix} = - \begin{bmatrix} \mathbf{M}_{1,1} & \cdots & \mathbf{M}_{1,N} \\ & \ddots & \vdots \\ & & \mathbf{M}_{N,N} \end{bmatrix} \begin{bmatrix} y_1^+ - y_0^+ \\ \vdots \\ y_{N-1}^+ - y_{N-2}^+ \\ y_N^+ - y_{N-1}^+ \end{bmatrix}, \quad (3.14)$$

where

$$\mathbf{M}_{i,j} = \begin{cases} A_i & \text{if } j = i, \\ A_i - A_{i+1} & \text{if } j \geq i+1. \end{cases} \quad (3.15)$$

The last equality of (3.7) can be written as

$$\frac{x_*^+ - x_N^+}{2} = y_N^+ - \frac{y_*^+ + y_0^+}{2}. \quad (3.16)$$

We can rewrite (3.14) using the inverse matrix $\mathbf{M}^{-1}$ as

$$\begin{bmatrix} y_1^+ - y_0^+ \\ \vdots \\ y_{N-1}^+ - y_{N-2}^+ \\ y_N^+ - y_{N-1}^+ \end{bmatrix} = - \begin{bmatrix} \mathbf{M}_{1,1}^{-1} & \cdots & \mathbf{M}_{1,N}^{-1} \\ & \ddots & \vdots \\ & & \mathbf{M}_{N,N}^{-1} \end{bmatrix} \begin{bmatrix} x_{N-1}^+ - x_{N-2}^+ \\ \vdots \\ x_1^+ - x_0^+ \\ x_0^+ - x_* \end{bmatrix}, \quad (3.17)$$

where

$$\mathbf{M}_{i,j}^{-1} = \begin{cases} 1/A_i & \text{if } j = i, \\ 1/A_i - 1/A_{i+1} & \text{if } j \geq i+1. \end{cases} \quad (3.18)$$

We set the multipliers $\lambda_{i,j}$ as

$$\lambda_{i,j} = \begin{cases} 1/A_{N-j+1} & \text{if } i = j-1, 1 \leq j \leq N, \\ 1/A_N & \text{if } i = *, j = 0, \\ 1/A_{N-j} - 1/A_{N-j+1} & \text{if } i = *, 1 \leq j \leq N-1, \\ 1 - 1/A_1 & \text{if } i = *, j = N, \\ 0 & \text{otherwise.} \end{cases} \quad (3.19)$$

Then, the condition (3.2) is equivalent to (3.16) and (3.17) with (3.18). Thus, (3.2) is equivalent to (3.7). It is straightforward to check (F1), (1.12), and that $\lambda_{i,j} \geq 0$ for all $i, j \in I_N^*$.

(ii) Let $A_k = 1/\sum_{j=0}^{N-k} \lambda_{*,j}$ for $k = 1, \dots, N$. Then, $\{A_k\}_{k=1}^N$ is a nondecreasing sequence in $[1, \infty)$. Using (1.12) and (F1), we can express all the multipliers $\lambda_{i,j}$ in terms of A_k as in (3.19). We set the multipliers $\nu_{i,j}$ as in (3.13). Reversing the argument of the proof for the case (i), one can see that this set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfies the desired properties. $\square$

The following theorem follows from Lemma 3.5, Theorems 3.1, and 3.3.

Theorem 3.6. *The following statements are equivalent:*

- (i) *There is a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfying (3.12), such that the inequality (3.11) holds for all [families $\{y_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.4)].*
- (ii) *There is a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfying (1.12), such that the inequality (3.1) holds for all [families $\{x_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.1)].*

Theorem 3.6 is valid for any $\mu \in [0, L]$. See [10, Thm. 1] for the same conclusion in the nonstrongly convex case ($\mu = 0$). See [11, Thm. 3] and [10, Thm. 2] for the essentially same conclusion in the corresponding continuous-time analysis ($\mu = 0$ and $1/L \rightarrow 0$).

3.5 Special correspondence that will connect Theorems 1.3 and 1.5

In this subsection, we present a variant of the (i) $\Rightarrow$ (ii) direction of Theorem 3.6 with a weaker assumption and a weaker conclusion. Precisely, we replace the condition (3.12) with the following one:

$$\nu_{i,j} = 0 \quad \text{for all } (i,j) \notin \{(k, k+1), (k+1, k), (N, k)\}_{0 \leq k \leq N-1} \cup \{(N, *)\}. \quad (3.20)$$

The following lemma is a variant of Lemma 3.5(i) in this context.

Lemma 3.7. *If a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfies (G1) and (3.20), then there is a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfying (F1) such that (3.2) is equivalent to (3.7).*

The proof of Lemma 3.7 follows a similar argument as in the proof of Lemma 3.5, but involves a tedious computation. In particular, we explicitly compute the multipliers $\lambda_{i,j}$ in terms of $\nu_{i,j}$. Thus, we defer the proof to Appendix A. By combining Lemma 3.7 with Theorems 3.1 and 3.3, we obtain the following result.

Theorem 3.8. *We have (i) $\Rightarrow$ (ii) for the following statements:*

- (i) *There is a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfying (3.20), such that the inequality (3.11) holds for all [families $\{y_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.4)].*
- (ii) *There is a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ such that the inequality (3.1) holds for all [families $\{x_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.1)].*

4 Proof of the “mirror” of the main theorem

In this section, we reprove Theorem 1.5. Precisely, we show that there is a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfying (3.20), such that (3.11) holds with

$$\begin{aligned} \tau &= L \max \left\{ \frac{1}{1 + \gamma L \sum_{j=1}^{2N} (1 - \gamma \mu)^{-j}}, (1 - \gamma L)^{2N} \right\} \\ &= L \max \left\{ \frac{1}{1 + \gamma L \cdot E_N(1 - \gamma \mu)}, \frac{1}{1 + \gamma L \cdot E_N(1 - \gamma L)} \right\} \end{aligned} \quad (4.1)$$

for all [families $\{y_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.4) with $\mathbf{H} = (\gamma L) \mathbf{I}_N$]. Throughout this section, we assume that the value of τ is always given by (4.1).

Note that if (3.20) holds, then (3.10) also holds. In this case, the inequality (3.11) can be equivalently written as

$$\begin{aligned} &\left(\frac{1}{2\tau} - \frac{1}{2L} \right) \|g_N\|^2 - \frac{\mu L}{2(L - \mu)} \left\| y_N - y_* - \frac{1}{L} g_N \right\|^2 - f_0 + f_N \\ &\leq \sum_{i,j \in I_N} \nu_{i,j} \left[f_j - f_i + \langle g_j, y_i - y_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2 \right. \\ &\quad \left. + \frac{\mu L}{2(L - \mu)} \left\| y_i - y_j - \frac{1}{L} (g_i - g_j) \right\|^2 \right]. \end{aligned}$$

Thus, when $\mu \geq 0$, a sufficient condition for (3.11) to hold is that the inequality

$$\begin{aligned} &\left(\frac{1}{2\tau} - \frac{1}{2L} \right) \|g_N\|^2 - f_0 + f_N \\ &\leq \sum_{i,j \in I_N} \nu_{i,j} \left[f_j - f_i + \langle g_j, y_i - y_j \rangle + \frac{1}{2L} \|g_i - g_j\|^2 \right. \\ &\quad \left. + \frac{\mu L}{2(L - \mu)} \left\| y_i - y_j - \frac{1}{L} (g_i - g_j) \right\|^2 \right] \end{aligned} \quad (4.2)$$

holds. Note that after substituting the value (4.1) of τ , we have

$$\frac{1}{2\tau} - \frac{1}{2L} = \frac{\gamma}{2} \min \{E_N(1 - \gamma\mu), E_N(1 - \gamma L)\}. \quad (4.3)$$

To remind the goal, we want to show that there is a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N}$ satisfying

$$\nu_{i,j} = 0 \quad \text{for all } (i,j) \notin \{(k,k+1), (k+1,k), (N,k)\}_{0 \leq k \leq N-1}, \quad (4.4)$$

such that the inequality (4.2) holds for all [families $\{y_i\}_{i \in I_N}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N}$ in $\mathbb{R}$ satisfying (2.4) with $\mathbf{H} = (\gamma L)\mathbf{I}_N$].

As in Theorem 3.3, we can derive two conditions that together imply this goal. The first condition, which plays the role of (G1), will be

$$\sum_{i \in I_N} \nu_{i,k} - \sum_{j \in I_N} \nu_{k,j} = \begin{cases} 1 & \text{if } k = N, \\ -1 & \text{if } k = 0, \\ 0 & \text{otherwise.} \end{cases} \quad (4.5)$$

The second condition can be formulated similarly to (G2), but we formulate it more explicitly in terms of the positive semidefiniteness of a certain matrix in the next subsection.

4.1 Semidefinite reformulation of PEP analysis

Note that the inequality (4.2) holds for all [families $\{y_i\}_{i \in I_N}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N}$ in $\mathbb{R}$ satisfying (2.4)] if and only if the inequality

$$\begin{aligned} & -\frac{1}{2L} \|g_0\|^2 + \frac{1}{2\tau} \|g_N\|^2 - f_0^+ + f_N^+ \\ & \leq \sum_{i,j \in I_N} \nu_{i,j} \left[f_j^+ - f_i^+ + \langle g_j, y_i^+ - y_j^+ \rangle + \frac{\mu L}{2(L - \mu)} \|y_i^+ - y_j^+\|^2 \right] \end{aligned} \quad (4.6)$$

holds for all [families $\{y_0\} \cup \{y_i^+\}_{i \in I_N}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N}$ in $\mathbb{R}^d$, and $\{f_i^+\}_{i \in I_N}$ in $\mathbb{R}$ satisfying (2.5)] (see Footnote 8). Define two $(N+1) \times (N+1)$ matrices $\mathbf{A}$ and $\mathbf{B}$ as

$$\mathbf{A} = \begin{bmatrix} 0 & b_0 & c_0 & c_0 & \cdots & c_0 & c_0 \\ 0 & a_1 & b_1 & c_1 & \cdots & c_1 & c_1 \\ 0 & 0 & a_2 & b_2 & \cdots & c_2 & c_2 \\ 0 & 0 & 0 & a_3 & \cdots & c_3 & c_3 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & a_{N-1} & b_{N-1} \\ 0 & 0 & 0 & 0 & \cdots & 0 & a_N \end{bmatrix},$$

where

$$\begin{aligned} a_k &= \nu_{k-1,k} \quad \text{for } k = 1, \dots, N, \\ b_k &= \begin{cases} -\nu_{k+1,k} - \nu_{N,k} & \text{if } k \leq N-2, \\ -\nu_{N,N-1} & \text{if } k = N-1, \end{cases} \\ c_k &= -\nu_{N,k} \quad \text{for } k = 0, \dots, N-2, \end{aligned}$$

and

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & d_1 & e_1 & e_1 & \cdots & e_1 & e_1 \\ 0 & e_1 & d_2 & e_2 & \cdots & e_2 & e_2 \\ 0 & e_1 & e_2 & d_3 & \cdots & e_3 & e_3 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & e_1 & e_2 & e_3 & \cdots & d_{N-1} & e_{N-1} \\ 0 & e_1 & e_2 & e_3 & \cdots & e_{N-1} & d_N \end{bmatrix},$$

where

$$d_k = \begin{cases} \nu_{k-1,k} + \nu_{k,k-1} + \sum_{j=0}^{k-1} \nu_{N,j} & \text{if } k \leq N-1, \\ \nu_{N-1,N} + \sum_{j=0}^{N-1} \nu_{N,j} & \text{if } k = N, \end{cases}$$

$$e_k = \sum_{j=0}^{k-1} \nu_{N,j} \quad \text{for } k = 0, \dots, N-1.$$

Then, we have

$$\sum_{i,j \in I_N} \nu_{ij} \langle g_j, y_i^+ - y_j^+ \rangle = -\langle \mathbf{g}, \mathbf{A}\mathbf{y} \rangle,$$

$$\sum_{i,j \in I_N} \nu_{ij} \|y_i^+ - y_j^+\|^2 = \langle \mathbf{B}\mathbf{y}, \mathbf{y} \rangle,$$

where $\mathbf{g} = [g_0, \dots, g_N]^T$ and $\mathbf{y} = [y_0^+ - y_0, y_1^+ - y_0^+, \dots, y_N^+ - y_{N-1}^+]^T$. Thus, when (4.5) holds, using $\mathbf{g} = -L\tilde{\mathbf{H}}^{-A}\mathbf{y}$, the inequality (4.6) can be equivalently written as

$$0 \leq S := \frac{1}{2L} \|g_0\|^2 - \frac{1}{2\tau} \|g_N\|^2 + \left\langle \left(L\mathbf{A}^T \tilde{\mathbf{H}}^{-A} + \frac{\mu L}{2(L-\mu)} \mathbf{B} \right) \mathbf{y}, \mathbf{y} \right\rangle. \quad (4.7)$$

Denote the first row and the last row of $\tilde{\mathbf{H}}^{-A}$ by $\mathbf{h}_0$ and $\mathbf{h}_N$, both written as column vectors. Then, we have $S = L \langle \mathbf{S}\mathbf{y}, \mathbf{y} \rangle$, where

$$\mathbf{S} = \mathbf{A}^T \tilde{\mathbf{H}}^{-A} + \frac{\kappa}{2(1-\kappa)} \mathbf{B} + \frac{1}{2} \mathbf{h}_0 \mathbf{h}_0^T - \frac{L}{2\tau} \mathbf{h}_N \mathbf{h}_N^T. \quad (4.8)$$

Thus, we obtain a condition which plays the role of (G2), namely the positive semidefiniteness of $\frac{1}{2}(\mathbf{S} + \mathbf{S}^T)$. We summarize the result in the following theorem.

Theorem 4.1. *Given a set of nonnegative multipliers $\{\nu_{i,j}\}_{i,j \in I_N}$ satisfying (4.4), the inequality (4.2) holds for all [families $\{y_i\}_{i \in I_N}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N}$ in $\mathbb{R}$ satisfying (2.4)] if the following two conditions are satisfied:*

(G1') *The condition (4.5) holds.*

(G2') *The matrix $\frac{1}{2}(\mathbf{S} + \mathbf{S}^T)$ is positive semidefinite, where $\mathbf{S}$ is defined in (4.8).*

4.2 Proof of Theorem 1.5

Given N , μ , and L (possibly with $\mu < 0$), the *optimal stepsize* $\gamma^*(N, \mu, L)$ is uniquely defined as the value of γ that optimizes the convergence rate (1.10). It is known that $\gamma^*(N, \mu, L)$ is the unique value of γ in $(0, 2/L)$ that makes the two arguments of the min function in (4.3) equal. It is also known that $E_N(\eta) < E_N(\rho)$ for $\gamma \in (0, \gamma^*(N, \mu, L))$ and $E_N(\eta) > E_N(\rho)$ for $\gamma \in (\gamma^*(N, \mu, L), 2/L)$. See [14, §2] for the mentioned facts. The following lemma shows the existence of a desired set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N}$ for the case $\gamma = \gamma^*(N, \mu, L)$.

Lemma 4.2. *When $\mu \in (-\infty, L)$ and $\gamma = \gamma^*(N, \mu, L)$, there is a set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N}$ satisfying (4.4) such that the inequality (4.2), with τ set as in (4.1), holds for all [families $\{y_i\}_{i \in I_N}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N}$ in $\mathbb{R}$ satisfying (2.4) with $\mathbf{H} = (\gamma L)\mathbf{I}_N$].*

In Appendix B, we prove Lemma 4.2 by explicitly setting the multipliers $\nu_{i,j}$ and then verifying the two conditions in Theorem 4.1. It is nice that once Lemma 4.2 is proved, its generalization (Lemma 4.3) follows effortlessly. The key idea is to consider γ and N as fixed, while allowing μ and L to vary.

Lemma 4.3. *Lemma 4.2 holds even when the assumption $\gamma = \gamma^*(N, \mu, L)$ is replaced by $\gamma \in (0, 2/L)$.*

Proof. Consider γ and N as fixed. Denote the set of multipliers given by Lemma 4.2 as $\{\nu_{i,j}\}_{i,j \in I_N}^{\mu,L}$, showing their dependency on μ and L .

Case 1. $\gamma = \gamma^*(N, \mu, L)$. This is proved by Lemma 4.2.

Case 2. $\gamma < \gamma^*(N, \mu, L)$. In this case, we have $E_N(1 - \gamma\mu) < E_N(1 - \gamma L)$, meaning that μ determines the convergence rate. By the intermediate value theorem, we can choose $L' \in (L, 2/\gamma)$ such that $E_N(1 - \gamma\mu) = E_N(1 - \gamma L')$ (see Table 1), or equivalently $\gamma = \gamma^*(N, \mu, L')$. Consider the set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N}^{\mu,L'}$ obtained from Lemma 4.2. Then, for all [families $\{y_i\}_{i \in I_N}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N}$ in $\mathbb{R}$ satisfying (2.4) with $\mathbf{H} = (\gamma L)\mathbf{I}_N$], the inequality (4.2), with L replaced by L' , holds. By applying Proposition 2.1(ii) to each summand in (4.2), we can conclude that the inequality (4.2) itself holds.

Case 3. $\gamma > \gamma^*(N, \mu, L)$. In this case, we have $E_N(1 - \gamma\mu) > E_N(1 - \gamma L)$, meaning that L determines the convergence rate. By the intermediate value theorem, we can choose $\mu' \in (-\infty, \mu)$ such that $E_N(1 - \gamma\mu') = E_N(1 - \gamma L)$ (see Table 1), or equivalently $\gamma = \gamma^*(N, \mu', L)$. Consider the set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N}^{\mu',L}$ obtained from Lemma 4.2. Then, for all [families $\{y_i\}_{i \in I_N}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N}$ in $\mathbb{R}$ satisfying (2.4) with $\mathbf{H} = (\gamma L)\mathbf{I}_N$], the inequality (4.2), with μ replaced by μ' , holds. By applying Proposition 2.1(i) to each summand in (4.2), we can conclude that the inequality (4.2) itself holds. $\square$

Proof of Theorem 1.5. Put $y_i \leftarrow x_i$ (see Footnote 5), $g_i \leftarrow \nabla f(x_i)$, and $f_i \leftarrow f(x_i)$ for all $i \in I_N^*$. Then, the inequality (4.2) holds by Lemma 4.3. As discussed at the beginning of this section, this implies that the inequality (3.11) holds. Since the right-hand side of this inequality is nonpositive by Proposition 1.1, we can conclude that the desired inequality (1.10) holds. $\square$

Remark 4.4. Throughout this section, the condition $\mu \geq 0$ was used only when showing that the inequality (4.2) implies the inequality (3.11). However, we can circumvent this step by using Proposition 2.2, which does not require $\mu \geq 0$. Thus, the proof of Theorem 1.5 remains valid even when $\mu < 0$.

5 Proof of the main theorem

We assume $\mu \geq 0$.

Proof of Theorem 1.3. For given N , μ , L , and γ , consider the set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N}$ given by Lemma 4.3. Incorporate the index $*$ by setting the multipliers $\nu_{i,*}$ and $\nu_{*,j}$ as in (3.10). Then, as discussed in the beginning of Section 4, the set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N^*}$ satisfies (3.20), and that the inequality (3.11), with τ set as in (4.1), holds for all [families $\{y_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.4) with $\mathbf{H} = (\gamma L)\mathbf{I}_N$]. By Theorem 3.8, there is a set of nonnegative multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ such that the inequality (3.1), with τ set as in (4.1), holds for all [families $\{x_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, $\{g_i\}_{i \in I_N^*}$ in $\mathbb{R}^d$, and $\{f_i\}_{i \in I_N^*}$ in $\mathbb{R}$ satisfying $g_* = 0$ and (2.1) with $\mathbf{H} = (\gamma L)\mathbf{I}_N$]. We now put $x_i \leftarrow x_i$ (see Footnote 5), $g_i \leftarrow \nabla f(x_i)$, and $f_i \leftarrow f(x_i)$ for all $i \in I_N^*$. Since the right-hand side of this inequality is nonpositive by Proposition 1.1, we can conclude that the desired inequality (1.9) holds. $\square$

Since our proofs of Lemmas 3.7 and 4.3 are constructive, we can explicitly compute the set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ used in the proof of Theorem 1.3.

Example 5.1. For $N = 5$, $\mu = 0.1$, $L = 1$, and $\gamma = \gamma^*(N, \mu, L)$, the set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ used in the proof of Theorem 1.3 is calculated as

$$[\lambda_{i,j}] = \begin{array}{c} \begin{array}{c} * \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} * \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} 0 \\ 0.0384 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \begin{array}{c} 1 \\ 0.0621 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \begin{array}{c} 2 \\ 0.1063 \\ 0.0119 \\ 0.0472 \\ 0 \\ 0 \\ 0 \end{array} \begin{array}{c} 3 \\ 0.1873 \\ 0.0060 \\ 0.0237 \\ 0.1186 \\ 0 \\ 0 \end{array} \begin{array}{c} 4 \\ 0.3342 \\ 0.0020 \\ 0.0080 \\ 0.0401 \\ 0.2876 \\ 0 \end{array} \begin{array}{c} 5 \\ 0.2718 \\ 0.0003 \\ 0.0013 \\ 0.0067 \\ 0.0479 \\ 0.6719 \end{array} \end{array}. \quad (5.1)$$

A MATLAB code for generating the multipliers $\lambda_{i,j}$ can be found in the file `generate_lambda.m` of the GitHub repository (see Section 1.4).

Remark 5.2. For the set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ we obtained, one can see that

$$\lambda_{i,j}/\lambda_{i',j} = \lambda_{i,j'}/\lambda_{i',j'} \quad (5.2)$$

holds whenever $0 \leq i, i' < j, j' \leq N$. This can be proved from the explicit formula for converting $\{\nu_{i,j}\}_{i,j \in I_N^*}$ into $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ provided in Appendix A. A similar property is satisfied by the conjectured set of multipliers in [7, Conj. 2].⁹

⁹In [7], it was conjectured that there is a set of multipliers $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ satisfying (5.2) whenever $i, i' < j-1, j'-1$, which proves Theorem 1.3 under a PEP analysis which is similar to that presented in Section 3.1. The structure of their $\{\lambda_{i,j}\}_{i,j \in I_N^*}$ is different from ours.

A Proof of Lemma 3.7

Let $A_k = 1 + \sum_{j=0}^{k-1} \nu_{N,j}$ for $k = 1, \dots, N$ and $B_k = \nu_{k,k-1}$ for $k = 1, \dots, N-1$. For convenience, let $B_N = 0$. Then, $\{A_k\}_{k=1}^N$ is a nondecreasing sequence in $[1, \infty)$ and $\{B_k\}_{k=1}^N$ is a sequence in $[0, \infty)$. Using (3.12) and (G1), we can express all the multipliers $\nu_{i,j}$ as

$$\nu_{i,j} = \begin{cases} A_j + B_j & \text{if } i = j - 1, 1 \leq j \leq N, \\ A_1 - 1 & \text{if } i = N, j = 0, \\ A_{j+1} - A_j & \text{if } i = N, 1 \leq j \leq N-1, \\ B_i & \text{if } j = i - 1, 1 \leq i \leq N-1, \\ 1 & \text{if } i = N, j = *, \\ 0 & \text{otherwise.} \end{cases}$$

Thus, (3.7) without the last equality can be expressed as (3.14) with

$$\mathbf{M}_{i,j} = \begin{cases} A_i + B_i & \text{if } j = i, \\ A_i - A_{i+1} - B_{i+1} & \text{if } j = i + 1, \\ A_i - A_{i+1} & \text{if } i + 2 \leq j \leq N. \end{cases}.$$

The last equality in (3.7) can be expressed as (3.16). The entries of the inverse matrix $\mathbf{M}^{-1}$ can be computed as

$$\mathbf{M}_{i,j}^{-1} = \begin{cases} \frac{1}{A_i + B_i} & \text{if } j = i, \\ \frac{1}{A_i + B_i} - \frac{1}{A_i + B_i} \frac{A_i}{A_{i+1} + B_{i+1}} & \text{if } j = i + 1, \\ \frac{1}{A_i + B_i} - \sum_{l=i+1}^j \left[\frac{1}{A_i + B_i} \frac{A_i}{A_{i+1} + B_{i+1}} \right. \\ \quad \left. \times \prod_{k=i+1}^{l-1} \frac{B_k}{A_{k+1} + B_{k+1}} \right] & \text{if } i + 2 \leq j \leq N. \end{cases} \quad (\text{A.1})$$

A MATLAB code for verifying (A.1) can be found in the file `inverse_transform.m` of the GitHub repository (see Section 1.4). We set the multipliers $\lambda_{i,j}$ as

$$\lambda_{i,j} = \begin{cases} \mathbf{M}_{N-j,N-i-1}^{-1} - \mathbf{M}_{N-j,N-i}^{-1} & \text{if } 0 \leq i < j \leq N-1, \\ \mathbf{M}_{N-j,N}^{-1} & \text{if } i = *, 0 \leq j \leq N-1, \\ \mathbf{M}_{N-i,N-i}^{-1} - \sum_{k=1}^{N-i-1} \left(\mathbf{M}_{k,N-i-1}^{-1} - \mathbf{M}_{k,N-i}^{-1} \right) & \text{if } 0 \leq i < j = N, \\ 1 - \sum_{l=0}^{N-1} \left[\mathbf{M}_{N-l,N-l}^{-1} \right. \\ \quad \left. - \sum_{k=1}^{N-l-1} \left(\mathbf{M}_{k,N-l-1}^{-1} - \mathbf{M}_{k,N-l}^{-1} \right) \right] & \text{if } i = *, j = N, \\ 0 & \text{otherwise.} \end{cases}$$

After substituting (A.1), we have

$$\lambda_{i,j} = \begin{cases} \frac{1}{A_{N-j}+B_{N-j}} \frac{A_{N-j}}{A_{N-j+1}+B_{N-j+1}} \times \prod_{k=N-j+1}^{N-i-1} \frac{B_k}{A_{k+1}+B_{k+1}} & \text{if } 0 \leq i < j \leq N-1, \\ \frac{1}{A_{N-j}+B_{N-j}} - \sum_{l=0}^{j-1} \left[\frac{1}{A_{N-j}+B_{N-j}} \times \frac{A_{N-j}}{A_{N-j+1}+B_{N-j+1}} \times \prod_{k=N-j+1}^{N-l-1} \frac{B_k}{A_{k+1}+B_{k+1}} \right] & \text{if } i = *, 0 \leq j \leq N-1, \\ \frac{1}{A_{N-i}+B_{N-i}} \prod_{k=1}^{N-i-1} \frac{B_k}{A_k+B_k} & \text{if } 0 \leq i < j = N, \\ 1 - \sum_{l=0}^{N-1} \left[\frac{1}{A_{N-l}+B_{N-l}} \prod_{k=1}^{N-l-1} \frac{B_k}{A_k+B_k} \right] & \text{if } i = *, j = N, \\ 0 & \text{otherwise.} \end{cases}$$

Then, the condition (3.2) is equivalent to (3.16) and (3.17) with (A.1). Thus, (3.2) is equivalent to (3.7). It is straightforward to check (F1).

It only remains to check $\lambda_{i,j} \geq 0$ for all $i, j \in I_N^*$. For the case where $0 \leq i < j \leq N-1$ and the case where $i = *$ and $0 \leq j \leq N-1$, this is clear. For the case where $i = *$ and $0 \leq j \leq N-1$, we have

$$\begin{aligned} \lambda_{i,j} &\geq \frac{1}{A_{N-j}+B_{N-j}} \\ &\quad - \sum_{l=0}^{j-1} \left(\frac{1}{A_{N-j}+B_{N-j}} \frac{A_{N-j}}{A_{N-j+1}+B_{N-j+1}} \prod_{k=N-j+1}^{N-l-1} \frac{B_k}{A_{N-j}+B_{k+1}} \right) \\ &= \frac{1}{A_{N-j}+B_{N-j}} \prod_{k=N-j+1}^N \frac{B_k}{A_{N-j}+B_k} \geq 0, \end{aligned}$$

where for the first inequality we used the fact that $A_k \geq A_{N-j}$ whenever $k \geq N-j$. For the case where $i = *$ and $j = N$, we have

$$\lambda_{i,j} \geq 1 - \sum_{l=0}^{N-1} \left[\frac{1}{1+B_{N-l}} \prod_{k=1}^{N-l-1} \frac{B_k}{1+B_k} \right] = \prod_{k=1}^N \frac{B_k}{1+B_k} \geq 0,$$

where for the first inequality we used the fact that $A_k \geq 1$ for all k . This completes the proof.

B Proof of Lemma 4.2

After choosing a set of multipliers $\{\nu_{i,j}\}_{i,j \in I_N}$, we verify the conditions (4.4), (4.5), $\nu_{i,j} \geq 0$ for all $i, j \in I$, and that the matrix $\frac{1}{2}(\mathbf{S} + \mathbf{S}^T)$ is positive semidefinite. The following propositions will be used throughout the proof.

Proposition B.1 ([14, Props. 2.12, 4.6]). *When $\gamma = \gamma^*(N, \mu, L)$, we have $\rho \in (-1, 0)$, $\eta \in (-\rho, \infty)$, $T_k(\rho, \eta) \geq 0$ for $k = 1, \dots, N-1$, and $T_N(\rho, \eta) = 0$.*

Proposition B.2. *For any $\rho \in (-1, 0)$, $\eta \in (0, \infty)$, and $N \in (0, \infty)$, the function*

$$\psi(t) = \begin{cases} \log \left(\frac{1+(1-\rho)(N+t)}{1+(1-\rho)(N-t)} \right) & \text{if } \eta = 1, \\ \log \left(\frac{-(\eta-\rho)+(1-\rho)\eta^{-t-N}}{-(\eta-\rho)+(1-\rho)\eta^{t-N}} \right) & \text{otherwise,} \end{cases}$$

is convex on $[0, N]$.

Proof. We present a proof for the case $\eta \neq 1$. The case $\eta = 1$ can be proved by applying similar reasoning. We need to show that the derivative

$$\frac{d\psi}{dt} = \frac{-(\log \eta)(1-\rho)\eta^{-t-N}}{-(\eta-\rho)+(1-\rho)\eta^{-t-N}} + \frac{-(\log \eta)(1-\rho)\eta^{t-N}}{-(\eta-\rho)+(1-\rho)\eta^{t-N}}$$

is nondecreasing on $[0, N]$. Note that if a differentiable function φ on $(-\infty, 0]$ is convex, then the function $t \mapsto \varphi(-N+t) + \varphi(-N-t)$ on $[0, N]$ is nondecreasing.¹⁰ Thus, it suffices to show that the function

$$\varphi(u) = \frac{-(\log \eta)(1-\rho)\eta^u}{-(\eta-\rho)+(1-\rho)\eta^u} = -(\log \eta) \left(1 + \frac{(\eta-\rho)}{-(\eta-\rho)+(1-\rho)\eta^u} \right)$$

is convex on $(-\infty, 0]$. Its second derivative can be calculated as

$$\frac{d^2\varphi}{du^2} = \frac{(\log \eta)^3 \eta^u (\eta-\rho)(1-\rho) (-(\eta-\rho)+(1-\rho)\eta^u) (-(\eta-\rho)-(1-\rho)\eta^u)}{(-(\eta-\rho)+(1-\rho)\eta^u)^4}.$$

It is straightforward to see that $\frac{d^2\varphi}{du^2}(u) \geq 0$ for all $u \in (-\infty, 0)$, meaning that φ is convex on $(-\infty, 0]$. This completes the proof. $\square$

B.1 Choosing multipliers

In this subsection, we set multipliers $\nu_{i,j}$. Since $\gamma = \gamma^*(N, \mu, L)$, we have $\rho, \eta \neq 0$ by Proposition B.1. Thus, $E_k(\eta)$ and $E_k(\rho)$ are finite for $k = 1, \dots, N$.

Case 1. $N = 1$. We set the multipliers $\nu_{i,j}$ as

$$\nu_{i,j} = \begin{cases} -\rho E_1(\rho) & \text{if } i = 1, j = 0, \\ 1 - \rho E_1(\rho) & \text{if } i = 0, j = 1. \end{cases}$$

Case 2. $N = 2$. Define two constants α_1 and β_1 as

$$\alpha_1 = \frac{T_1(\rho, \eta)}{F_1(\eta)}, \quad \beta_1 = \frac{\eta - \rho}{\eta} E_1(\rho) - \frac{T_1(\rho, \eta)}{F_1(\eta)}.$$

¹⁰We have $\frac{d}{dt} \{\varphi(-N+t) + \varphi(-N-t)\} = \frac{d\varphi}{dt}(-N+t) - \frac{d\varphi}{dt}(-N-t) \geq 0$, where the inequality follows from the fact that $\frac{d\varphi}{dt}$ is nondecreasing.

We set the multipliers $\nu_{i,j}$ as

$$\nu_{i,j} = \begin{cases} -\frac{\eta\rho}{\eta-\rho}\beta_1 & \text{if } i = 1, j = 0, \\ -\frac{\eta\rho}{\eta-\rho}\alpha_1 & \text{if } i = 2, j = 0, \\ 1 - \frac{\eta\rho}{\eta-\rho}(\alpha_1 + \beta_1) & \text{if } i = 0, j = 1, \\ -\rho E_2(\rho) + \frac{1+\rho}{\eta-\rho}T_1(\rho, \eta) & \text{if } i = 2, j = 1, \\ 1 - \frac{\eta\rho}{\eta-\rho}\alpha_1 - \rho E_2(\rho) + \frac{1+\rho}{\eta-\rho}T_1(\rho, \eta) & \text{if } i = 1, j = 2, \\ 0 & \text{otherwise.} \end{cases}$$

Case 3. $N \geq 3$. Define two sequences $\{\alpha_k\}_{k=1}^{N-1}$ and $\{\beta_k\}_{k=1}^{N-1}$ as

$$\alpha_k = \begin{cases} \frac{T_1(\rho, \eta)}{F_{N-1}(\eta)} & \text{if } k = 1, \\ -\frac{1}{\rho} \frac{T_1(\rho, \eta)}{F_{N-1}(\eta)} + \left(\frac{T_2(\rho, \eta)}{F_{N-2}(\eta)} - \frac{T_1(\rho, \eta)}{F_{N-1}(\eta)} \right) & \text{if } k = 2, \\ -\frac{1}{\rho} \left(\frac{T_{k-1}(\rho, \eta)}{F_{N-k+1}(\eta)} - \frac{T_{k-2}(\rho, \eta)}{F_{N-k+2}(\eta)} \right) + \left(\frac{T_k(\rho, \eta)}{F_{N-k}(\eta)} - \frac{T_{k-1}(\rho, \eta)}{F_{N-k+1}(\eta)} \right) & \text{if } 3 \leq k \leq N-1, \end{cases}$$

and

$$\beta_k = \frac{\eta - \rho}{\eta} E_k(\rho) - \frac{T_k(\rho, \eta)}{F_{N-k}(\eta)} \quad \text{for } k = 1, \dots, N-1.$$

We set the multipliers $\nu_{i,j}$ as

$$\nu_{i,j} = \begin{cases} -\frac{\eta\rho}{\eta-\rho}\beta_{j+1} & \text{if } i = j+1, 0 \leq j \leq N-2, \\ -\frac{\eta\rho}{\eta-\rho}\alpha_{j+1} & \text{if } i = N, 0 \leq j \leq N-2, \\ 1 - \frac{\eta\rho}{\eta-\rho} \left(\sum_{k=1}^j \alpha_k + \beta_j \right) & \text{if } i = j-1, 1 \leq j \leq N-1, \\ \frac{\eta}{\eta-\rho} \left(\frac{T_{N-1}(\rho, \eta)}{F_1(\eta)} - \frac{T_{N-2}(\rho, \eta)}{F_2(\eta)} \right) - \rho E_N(\rho) + \frac{\eta\rho}{\eta-\rho} \frac{T_{N-1}(\rho, \eta)}{F_1(\eta)} & \text{if } i = N, j = N-1, \\ \frac{\eta}{\eta-\rho} \left(\frac{T_{N-1}(\rho, \eta)}{F_1(\eta)} - \frac{T_{N-2}(\rho, \eta)}{F_2(\eta)} \right) - \rho E_N(\rho) + \frac{\eta\rho}{\eta-\rho} \frac{T_{N-1}(\rho, \eta)}{F_1(\eta)} - \frac{\eta\rho}{\eta-\rho} \sum_{k=1}^{N-1} \alpha_k + 1 & \text{if } i = N-1, j = N, \\ 0 & \text{otherwise.} \end{cases}$$

It is clear that the conditions (4.4) and (4.5) hold for any case.

B.2 Verifying nonnegativity of multipliers

For simplicity, we only provide a proof for the third case in the previous subsection. The second case is similar. The first case is trivial. We only need to show that α_k , β_k , and $\nu_{N,N-1}$ are nonnegative, since every multiplier $\nu_{i,j}$ is a weighted sum of these plus a nonnegative constant.

B.2.1 Verifying nonnegativity of α_k

For $k = 1, \dots, N-1$, the nonnegativity of α_k follows from Proposition B.1 and the following proposition.

Proposition B.3. *The inequality*

$$\frac{T_{k+1}(\rho, \eta)}{F_{N-k-1}(\eta)} - \frac{T_k(\rho, \eta)}{F_{N-k}(\eta)} \geq 0 \quad (\text{B.1})$$

holds for $k = 1, \dots, N-2$

Proof. We consider two cases: $\eta \in (0, 1]$ and $\eta \in (1, \infty)$.

Case 1. $\eta \in (0, 1]$. Define a sequence $\{\varphi_k\}_{k=1}^N$ by

$$\begin{aligned} \varphi_k &= \eta^{2k-2} (\eta^{-2k+1} + \eta^{-2k} - \rho^{-2k+1} - \rho^{-2k}) \\ &= (\eta^{-1} + \eta^{-2}) - \frac{\eta^{2k-2}}{\rho^{2k-2}} (\rho^{-1} + \rho^{-2}). \end{aligned}$$

Then, φ_k is nonincreasing by Proposition B.1, and we have

$$T_k(\rho, \eta) = \varphi_1 + \eta^{-2}\varphi_2 + \dots + \eta^{-2k+2}\varphi_k.$$

Thus, for $k = 1, \dots, N-2$, we have

$$\begin{aligned} 0 &= T_N(\rho, \eta) \\ &= \varphi_1 + \eta^{-2}\varphi_2 + \dots + \eta^{-2N+2}\varphi_N \\ &\leq \varphi_1 + \eta^{-2}\varphi_2 + \dots + \eta^{-2k+2}\varphi_k \\ &\quad + (\eta^{-2k} + \eta^{-2k-2} \dots + \eta^{-2N+2}) \varphi_{k+1} \\ &= T_k(\rho, \eta) + (\eta^{-2k} + \eta^{-2k-2} + \dots + \eta^{-2N+2}) \varphi_{k+1}. \end{aligned}$$

Using $T_{k+1}(\rho, \eta) = T_k(\rho, \eta) + \eta^{-2k}\varphi_{k+1}$, we have

$$\begin{aligned} 0 &\leq T_k(\rho, \eta) + (\eta^{-2k} + \eta^{-2k-2} \dots + \eta^{-2N+2}) \eta^{2k} (T_{k+1}(\rho, \eta) - T_k(\rho, \eta)) \\ &= (1 + \eta^{-2} + \dots + \eta^{-2N+2k+2}) T_{k+1}(\rho, \eta) - (\eta^{-2} + \dots + \eta^{-2N+2k+2}) T_k(\rho, \eta). \end{aligned}$$

Thus, we have

$$\frac{T_{k+1}(\rho, \eta)}{T_k(\rho, \eta)} \geq \frac{\eta^{-2} + \dots + \eta^{-2N+2k+2}}{1 + \eta^{-2} + \dots + \eta^{-2N+2k+2}}.$$

To prove (B.1), we only need to show

$$\frac{\eta^{-2} + \dots + \eta^{-2N+2k+2}}{1 + \eta^{-2} + \dots + \eta^{-2N+2k+2}} \geq \frac{\eta + \eta^2 + \dots + \eta^{N-k-1}}{\eta + \eta^2 + \dots + \eta^{N-k}}.$$

One can verify that this inequality holds for all $\eta \in (0, 1]$.

Case 2. $\eta \in (1, \infty)$. Define a sequence $\{\varphi_k\}_{k=1}^N$ by

$$\varphi_k = \eta^{-2k+1} + \eta^{-2k} - \rho^{-2k+1} - \rho^{-2k}.$$

Then, φ_k is nonincreasing by Proposition B.1, and we have

$$T_k(\rho, \eta) = \varphi_1 + \varphi_2 + \dots + \varphi_k.$$

Following a similar argument as in the first case, we arrive at

$$\frac{T_{k+1}(\rho, \eta)}{T_k(\rho, \eta)} \geq \frac{N - k - 1}{N - k}.$$

Thus, we only need to show

$$\frac{N - k - 1}{N - k} \geq \frac{\eta + \eta^2 + \dots + \eta^{N-k-1}}{\eta + \eta^2 + \dots + \eta^{N-k}}.$$

One can verify that this inequality holds for all $\eta \in (1, \infty)$. $\square$

B.2.2 Verifying nonnegativity of β_k

To show $\beta_k \geq 0$ for $k = 1, \dots, N - 1$, it suffices to prove the following proposition.

Proposition B.4. *The inequality*

$$\frac{\eta - \rho}{\eta} E_k(\rho) - \frac{T_k(\rho, \eta)}{F_{N-k}(\eta)} \geq 0 \quad (\text{B.2})$$

holds for $k = 1, \dots, N - 1$.

Proof. We present a proof for the case where $\eta \neq 1$. The case where $\eta = 1$ can be handled using the same argument with the expressions $E_k(\eta) = 2k$ and $F_{N-k}(\eta) = N - k$. Substituting $T_k(\rho, \eta) = E_k(\eta) - E_k(\rho)$ and dividing both sides by $\frac{\eta - \rho}{\eta} + \frac{1}{F_{N-k}(\eta)}$, the inequality (B.2) can be equivalently written as

$$E_k(\rho) - \frac{\eta}{(\eta - \rho)F_{N-k}(\eta) + \eta} E_k(\eta) \geq 0. \quad (\text{B.3})$$

Using the expressions $E_k(x) = \frac{x^{-2k}-1}{1-x}$ and $F_k(x) = \frac{x(1-x^k)}{1-x}$, this inequality can be equivalently written as

$$\frac{(-\rho)^{-2k} - 1}{1 - \rho} \geq \frac{\eta^{-2k} - 1}{(\eta - \rho)(1 - \eta^{N-k}) + (1 - \eta)}. \quad (\text{B.4})$$

Multiplying both sides by $1 - \rho$ and adding 1 to each side, we obtain

$$\begin{aligned} (-\rho)^{-2k} &\geq \frac{(1 - \rho)(\eta^{-2k} - 1)}{(\eta - \rho)(1 - \eta^{N-k}) + (1 - \eta)} + 1 \\ &= \frac{-(\eta - \rho) + (1 - \rho)\eta^{-k-N}}{-(\eta - \rho) + (1 - \rho)\eta^{k-N}}. \end{aligned}$$

By taking logarithms, we obtain

$$-2k \log(-\rho) \geq \log \left(\frac{-(\eta - \rho) + (1 - \rho)\eta^{-k-N}}{-(\eta - \rho) + (1 - \rho)\eta^{k-N}} \right).$$

We consider k as a real variable, and view both sides as functions of k . This inequality becomes an equality at $k = 0$ and $k = N$, as is clear from its equivalent forms (B.3) and (B.4). By Proposition B.2, the right-hand side is convex on $[0, N]$. Therefore, the given inequality is valid for all $k \in [0, N]$. $\square$

B.2.3 Verifying nonnegativity of $\nu_{N,N-1}$

We rewrite $\nu_{N,N-1}$ as

$$\begin{aligned}\nu_{N,N-1} &= -\rho E_N(\rho) + \frac{\rho}{\eta - \rho} T_{N-1}(\rho, \eta) + \frac{\eta}{\eta - \rho} \left(\frac{T_{N-1}(\rho, \eta)}{\eta} - \frac{T_{N-2}(\rho, \eta)}{\eta + \eta^2} \right) \\ &= -\frac{\rho\eta}{\eta - \rho} \left(\frac{\eta - \rho}{\eta} E_{N-1}(\rho) - \frac{T_{N-1}(\rho, \eta)}{F_1(\eta)} \right) \\ &\quad + \frac{\eta}{\eta - \rho} \left(\frac{T_{N-1}(\rho, \eta)}{F_1(\eta)} - \frac{T_{N-2}(\rho, \eta)}{F_2(\eta)} \right) - \rho(\rho^{-2N+1} + \rho^{-2N}).\end{aligned}$$

The nonnegativity of $\nu_{N,N-1}$ now follows from Propositions B.1, B.3, and B.4.

B.3 Verifying positive semidefiniteness of $\frac{1}{2}(\mathbf{S} + \mathbf{S}^T)$

From $\mathbf{H} = (\gamma L)\mathbf{I}_N = (1 - \rho)\mathbf{I}_N$, we obtain

$$\tilde{\mathbf{H}}^{-A} = \begin{bmatrix} 1 & & & & \\ \rho & 1 & & & \\ \rho^2 & \rho & 1 & & \\ \vdots & \vdots & \vdots & \ddots & \\ \rho^N & \rho^{N-1} & \rho^{N-2} & \dots & 1 \end{bmatrix}.$$

Since $E_N(\eta) = E_N(\rho)$, we have $\tau = L\rho^{2N}$. We substitute all expressions into (4.8), and then find a nice expression for the matrix $\frac{1}{2}(\mathbf{S} + \mathbf{S}^T)$. This task is summarized in the following proposition.

Proposition B.5. *With $\mathbf{H} = (1 - \rho)\mathbf{I}_N$, $\tau = L\rho^{2N}$, and the multipliers $\nu_{i,j}$ chosen in Appendix B.1, the matrix $\frac{1}{2}(\mathbf{S} + \mathbf{S}^T)$ in Section 4.1 can be expressed as*

$$\frac{1}{2}(\mathbf{S} + \mathbf{S}^T) = \begin{cases} \frac{\eta^2(1-\rho)}{2(\eta-\rho)^2} \begin{bmatrix} 0 & 0 \\ 0 & T_1(\rho, \eta) \end{bmatrix} & \text{if } N = 1, \\ \frac{\eta^2(1-\rho)}{2(\eta-\rho)^2} \sum_{k=1}^N \delta_k \mathbf{v}_k \mathbf{v}_k^T & \text{if } N \geq 2, \end{cases} \quad (\text{B.5})$$

where the sequence $\{\delta_k\}_{k=1}^N$ is defined by

$$\delta_k = \begin{cases} T_1(\rho, \eta) & \text{if } k = 1, \\ T_k(\rho, \eta) - \frac{F_{N-k}(\eta)^2}{F_{N-k+1}(\eta)^2} T_{k-1}(\rho, \eta) & \text{if } 2 \leq k \leq N, \end{cases}$$

and the vectors $\mathbf{v}_1, \dots, \mathbf{v}_N \in \mathbb{R}^{N+1}$ are defined by

$$\begin{aligned}\mathbf{v}_1 &= [0, 1, -1/F_{N-1}(\eta), \dots, -1/F_{N-1}(\eta)]^T, \\ \mathbf{v}_2 &= [0, 0, 1, -1/F_{N-2}(\eta), \dots, -1/F_{N-2}(\eta)]^T, \\ &\vdots \\ \mathbf{v}_{N-1} &= [0, \dots, 0, 1, -1/F_1(\eta)]^T, \\ \mathbf{v}_N &= [0, \dots, 0, 1]^T.\end{aligned}$$

A MATLAB code for verifying Proposition B.5 can be found in the file `pep_matrix.m` of the GitHub repository (see Section 1.4). Due to this nice expression, proving the positive semidefiniteness of $\frac{1}{2}(\mathbf{S} + \mathbf{S}^T)$ reduces to proving $\delta_k \geq 0$ for $k = 1, \dots, N$. This follows from $\frac{F_{N-k}(\eta)}{F_{N-k+1}(\eta)} \in [0, 1]$, Propositions B.1, and B.3.

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